

Assam Power Sector

August 2023



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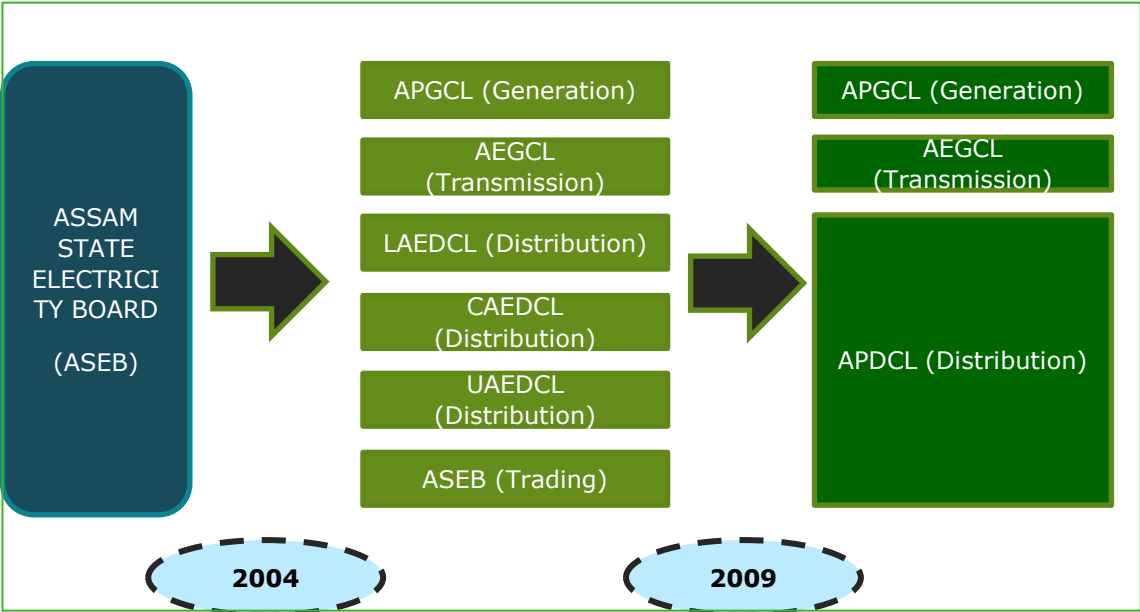
1. Assam Power Sector Overview



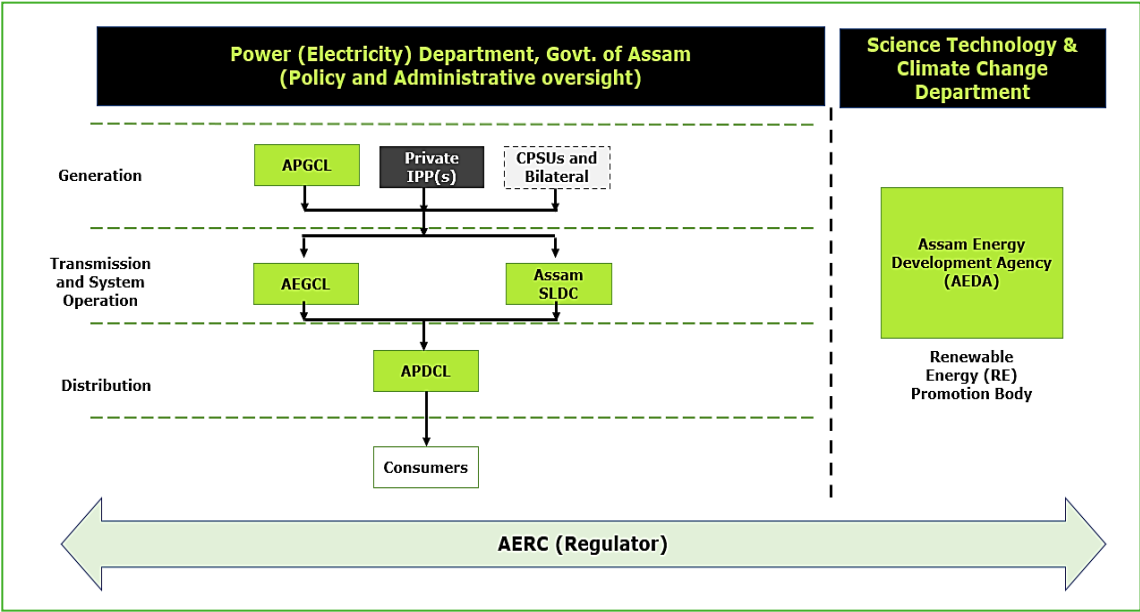
Assam Power Sector Profile

Assam power sector entities restructured in 2004 with unbundling of Assam SEB and followed by merging of multiple Distribution utilities and Trading company into one state Discom in 2009

Power sector Restructuring



Current Framework



2004 (Unbundling of ASEB)

The State's power sector went through major reforms in 2004, wherein the erstwhile Assam State Electricity Board (ASEB) was restructured into following five entities to carry out functions of generation, transmission and distribution of electricity

- ❑ Assam Power Generation Corporation Limited (APGCL) (Generating Company)
- ❑ Assam Electricity Grid Corporation Limited (AEGCL) (Transmission Company)
- ❑ Lower, Central and Upper Assam Distribution Company Limited (Three Electricity Distribution Companies)

2009 (Formation of APDCL)

- ❑ CAEDCL & UAEDCL Distribution Company merged with the LAEDCL to form a single distribution company viz. Assam Power Distribution Company (APDCL)

Generation Capacity - Current

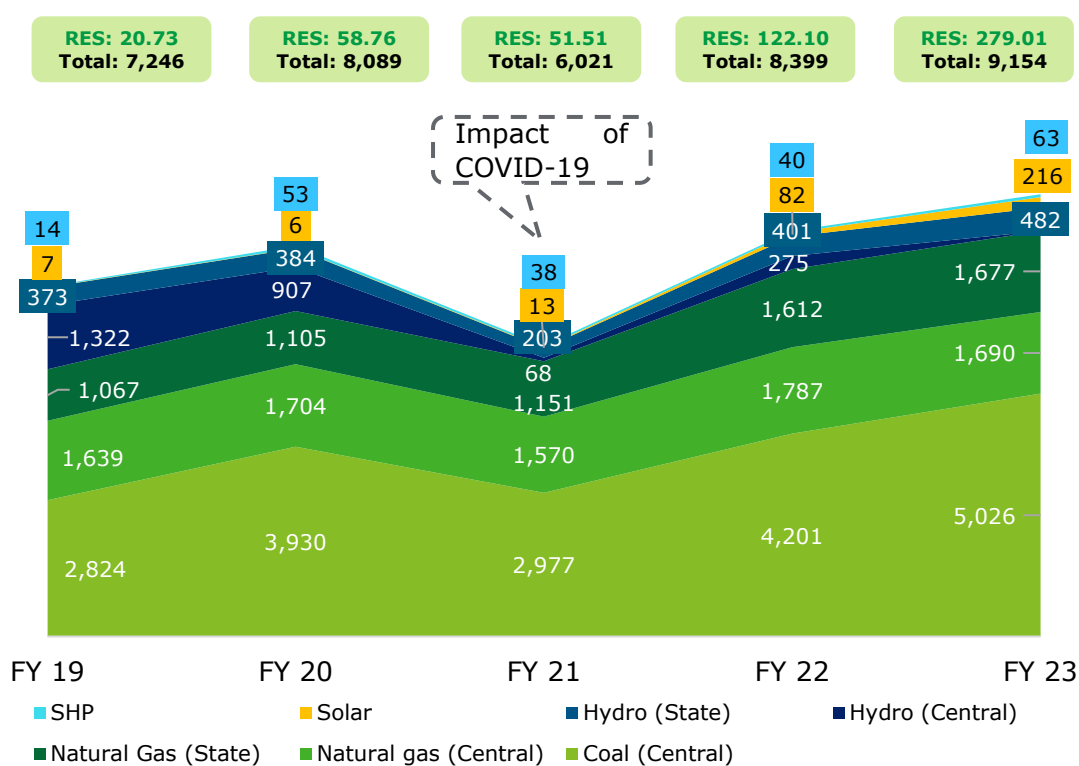
Assam State-owned generation asset portfolio is heavily skewed towards Natural Gas based projects (~74% in FY 23)

Installed Capacity in Assam (MW)

	FY 19	FY 20	FY 21	FY 22	FY 23	CAGR
Coal (Central)	750	750	750	750	750	0%
Coal (State)	0	0	0	0	0	-
Natural gas (Central)	291	291	291	291	291	0%
Natural Gas (Pvt)	25	25	25	0	0	-100%
Natural Gas (State)	328	328	329	329	306	-2%
Hydro (Central)	250	250	250	250	250	0%
Hydro (State)	100	100	100	100	100	0%
RES (Central)	25	25	25	25	25	0%
RES (State)	10.5	10.5	10.5	13.5	13.5	0%
RES (Pvt.)	27	45	49	124	154	55%
Total	1,807	1,825	1,830	1,883	1,890	1.13%

- Central Sector projects contribute 70% of installed capacity followed by state sector (22%)
- High reliance on Coal (40%) and Gas projects (32%)
- RE + Large Hydro constitutes 29% of Capacity
- Private RE capacity addition ~55% CAGR from FY 19 to FY 23

Energy Generation in Assam (MUs)

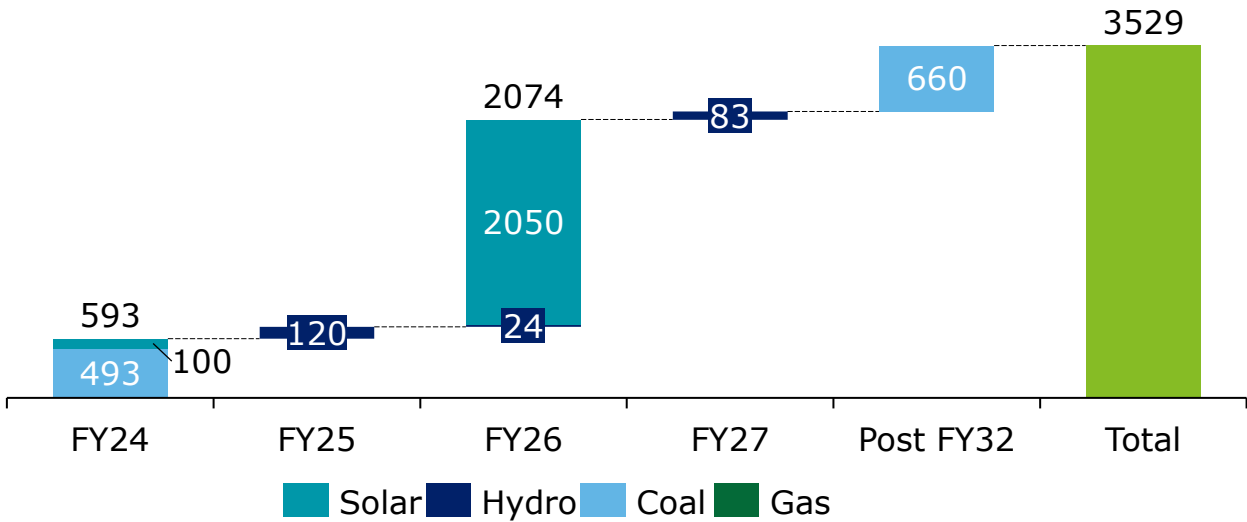


- Central plants are not fully allocated to Assam - Centrally owned 750 MW Bongaigaon TPS in Assam has ~50% allocation to the state
- Natural Gas based generation share is 35-37%
- RE+ Hydro contributes 8% of energy generation

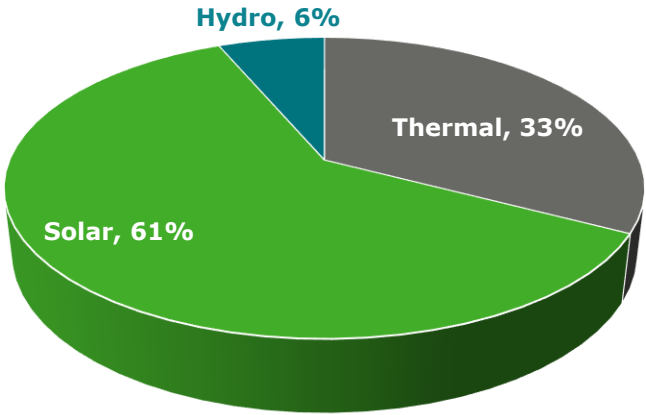
Generation Capacity - Additions

Around 3.5 GW of projects are currently in the pipeline

Proposed Projects (MW)



Resource wise share in pipeline (%)

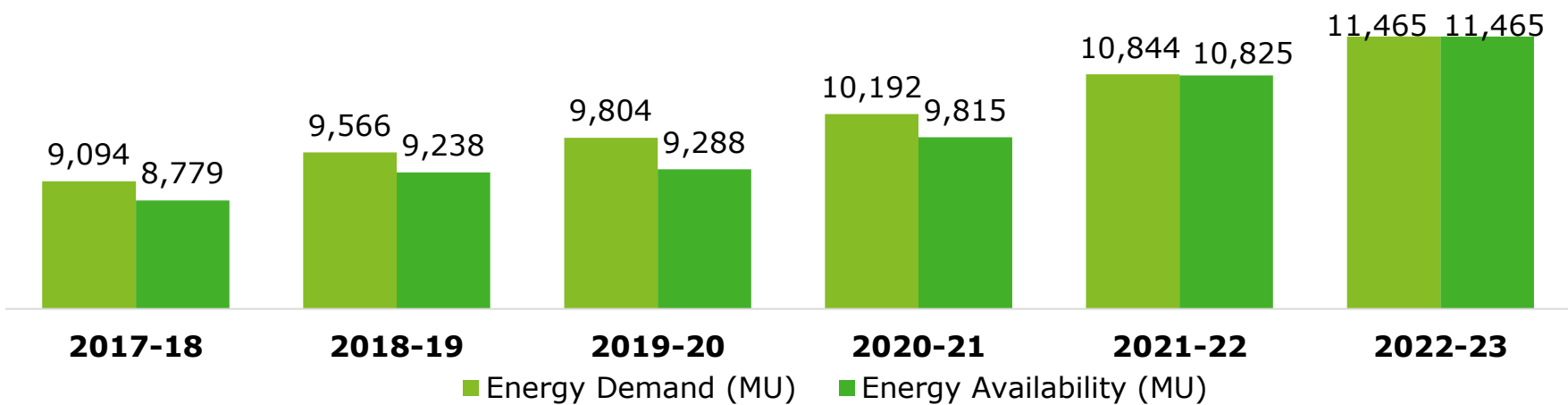


S No	Name of project	Capacity (MW)	Type	Agency	Expected COD
1	NUPPL project (3 X 660 MW)	493	Thermal	APDCL	2023-24
2	Namrup Solar PV	20	Solar	APGCL	2023-24
3	Sonabeel Floating Solar	60	Solar	APGCL	2023-24
4	Majuli Solar PV Project	20	Solar	APGCL	2023-24
5	Lower Kopili HEP	120	Hydro	APGCL	2024-25
6	Borpani Middle II SHEP	24	Hydro	APGCL	2025-26
7	Floating solar – PPP	100	Solar	APDCL	By 2025-26
8	Solar project - CPSU Ph-II Sch.	10	Solar	APDCL	By 2025-26
9	Solar project – PPP	50	Solar	APDCL	By 2025-26
10	Solar project – PPP	70	Solar	APDCL	By 2025-26
11	Solar project – PPP	200	Solar	APDCL	By 2025-26
12	JV b/w APDCL and NLCIL	1,000	Solar	APDCL	By 2025-26
13	JV b/w APGCL & Oil India	620	Solar	APDCL	By 2025-26
14	Borpani Middle I SHEP	23	Hydro	APGCL	2026-27
15	Borpani Upper Stage SHEP	60	Hydro	APGCL	2026-27
16	Margherita Thermal Project	660	Thermal	APGCL	Beyond 2032
TOTAL (MW)		3,529			

Demand Supply Scenario in Assam

Assam has been slightly ahead of the curve with respect to rate of growth in Energy Demand, in comparison to All India numbers and significantly ahead with respect to North-east region as a whole

Energy Demand and Availability in Assam (MU)



FY 2017-18 to FY 22-23

Assam:

Energy Demand CAGR	4.7%
Energy Availability CAGR	5.5%

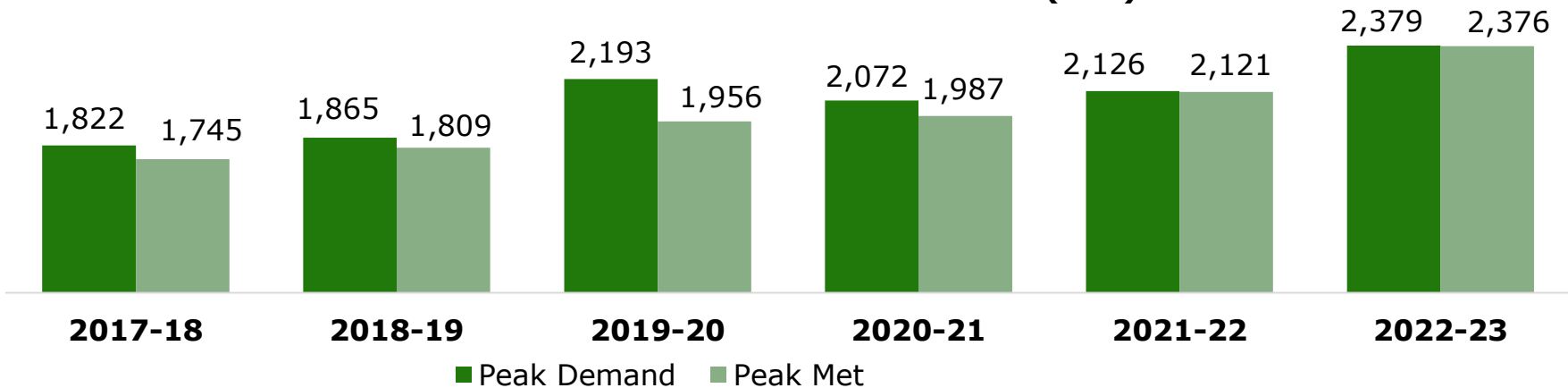
NE Region:

Energy Demand CAGR	3.3%
Energy Availability CAGR	4.5%

All India:

Energy Demand CAGR	4.5%
Energy Availability CAGR	4.5%

Peak Demand and Peak Met in Assam (MW)



FY 2017-18 to FY 22-23

Assam:

Peak Demand CAGR	5.5%
Peak Met CAGR	6.4%

NE Region:

Peak Demand CAGR	5.5%
Peak Met CAGR	6.4%

All India:

Peak Demand CAGR	5.6%
Peak Met CAGR	5.2%

- Steady increase in Energy and Peak Demand in Assam; highest in the North-East Region and even higher than National growth rate (in terms of energy demand)
- Energy requirement has grown at CAGR of 8.59%; projected to grow at CAGR of 6.44%

Energy Balance

Significant increase in Energy requirement and Peak Demand envisaged for Assam based on APDCL True Up Order and CEA Projections

Demand Projections based on 20 th EPS trends									
Assam	FY 2024	FY 2025	FY 2026	FY 2027	FY 2028	FY 2029	FY 2030	FY 2031	FY 2032
Energy Requirement (MU)	13,124	13,969	14,868	15,825	16,844	17,929	19,083	20,312	21,619
T&D Losses	16.78%	16.41%	16.06%	15.70%	15.36%	15.03%	14.70%	14.38%	14.06%
Energy Supply (MU)	10,922	11,676	12,481	13,340	14,257	15,235	16,278	17,391	18,579
Peak Demand (MW)	2,661	2,841	3,032	3,237	3,455	3,687	3,935	4,198	4,478
Average Demand (MW)	1,498	1,595	1,697	1,807	1,923	2,047	2,178	2,319	2,468

- Steady increase in Energy and Peak Demand in Assam; highest in the North-East Region and even higher than National growth rate (in terms of energy demand)
- Energy requirement has grown at CAGR of 8.59%; projected to grow at CAGR of 6.44%

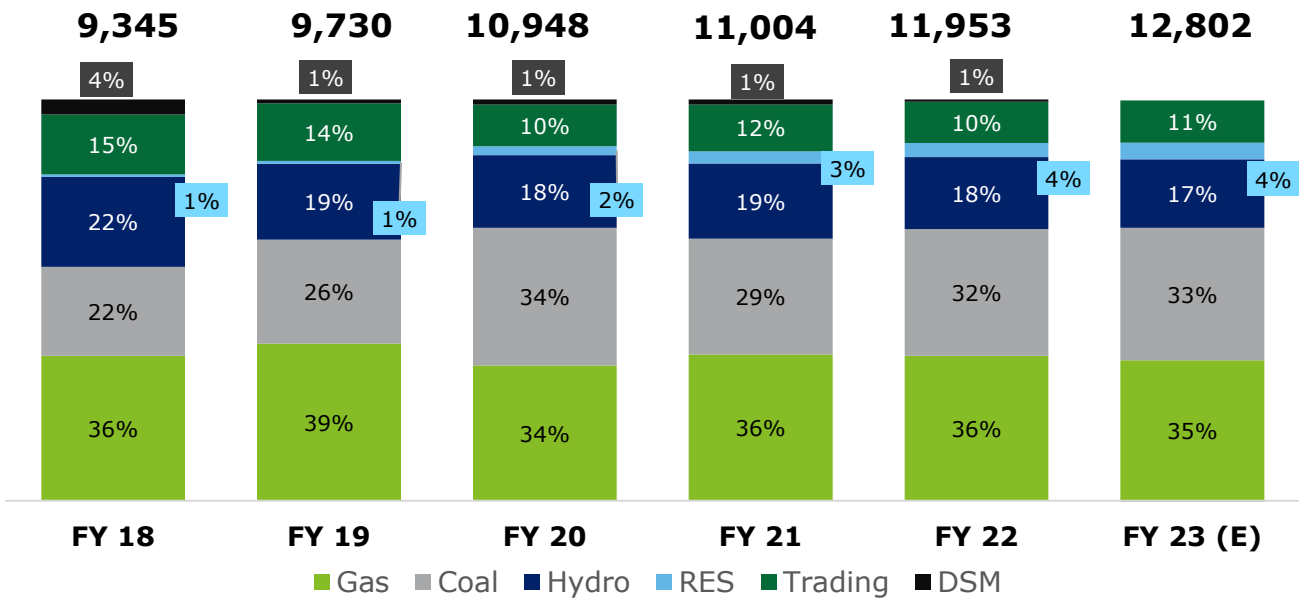
APDCL Power Purchase Mix

Coal and gas-based power procurement has dominated the power procurement portfolio of APDCL over the years; major dependence on power from Centrally-owned assets and Short-Term solutions (combined ~82% in FY 23) to cater to power requirement of state

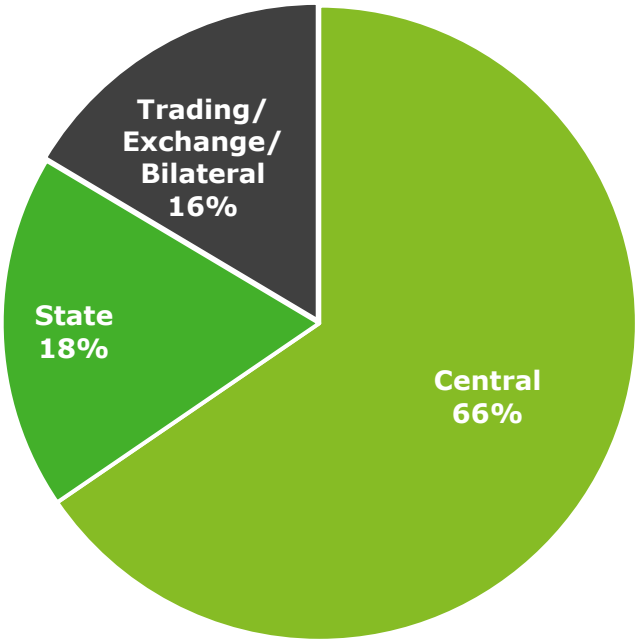
APDCL Energy Purchase Mix (%)

Net power purchase (MUs):

CAGR: 6.5%



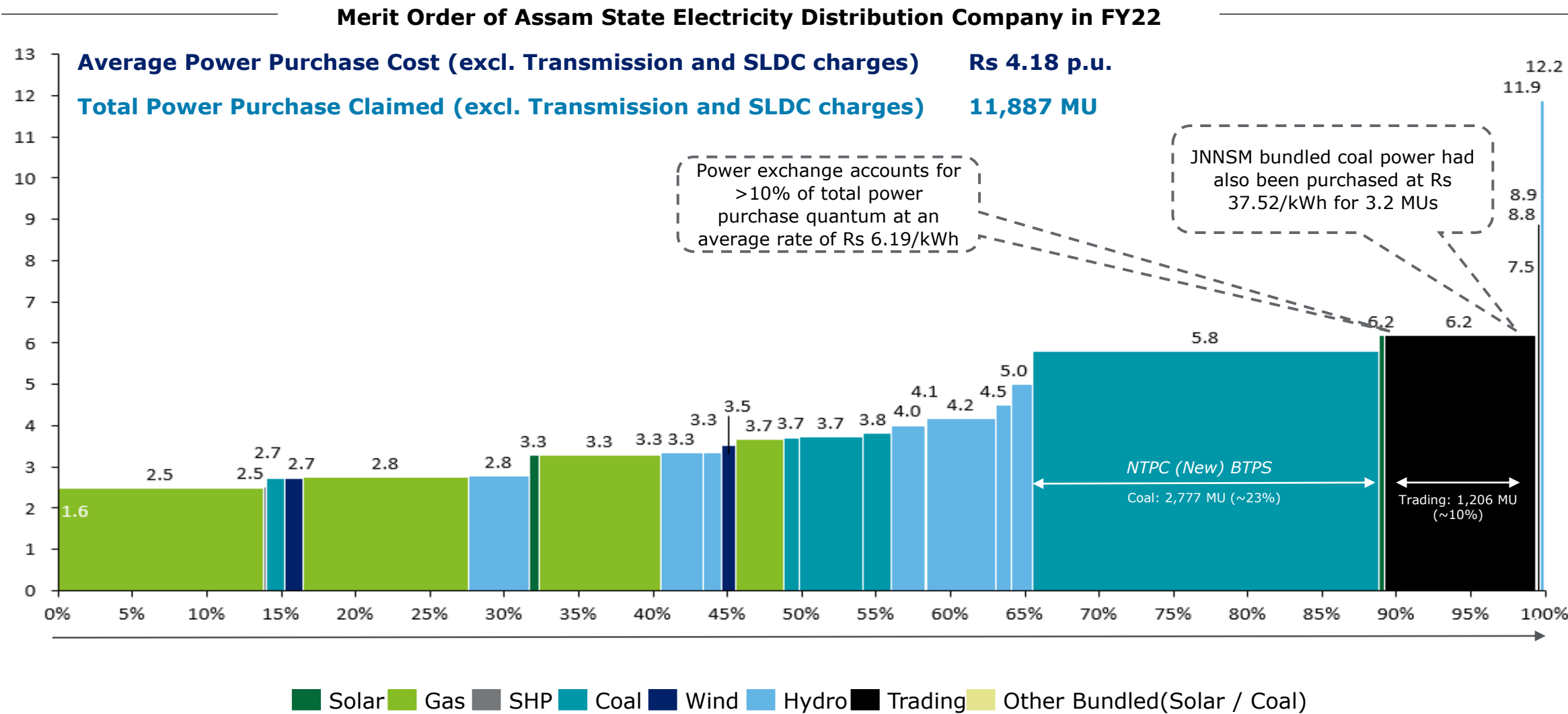
Allocation-wise Purchase Mix FY 23 (%)



State-owned assets accounted for only 18% in terms of power procurement by APDCL in FY 23. Therefore, greater reliance on open market operations leading to higher power purchase costs by Assam DISCOM

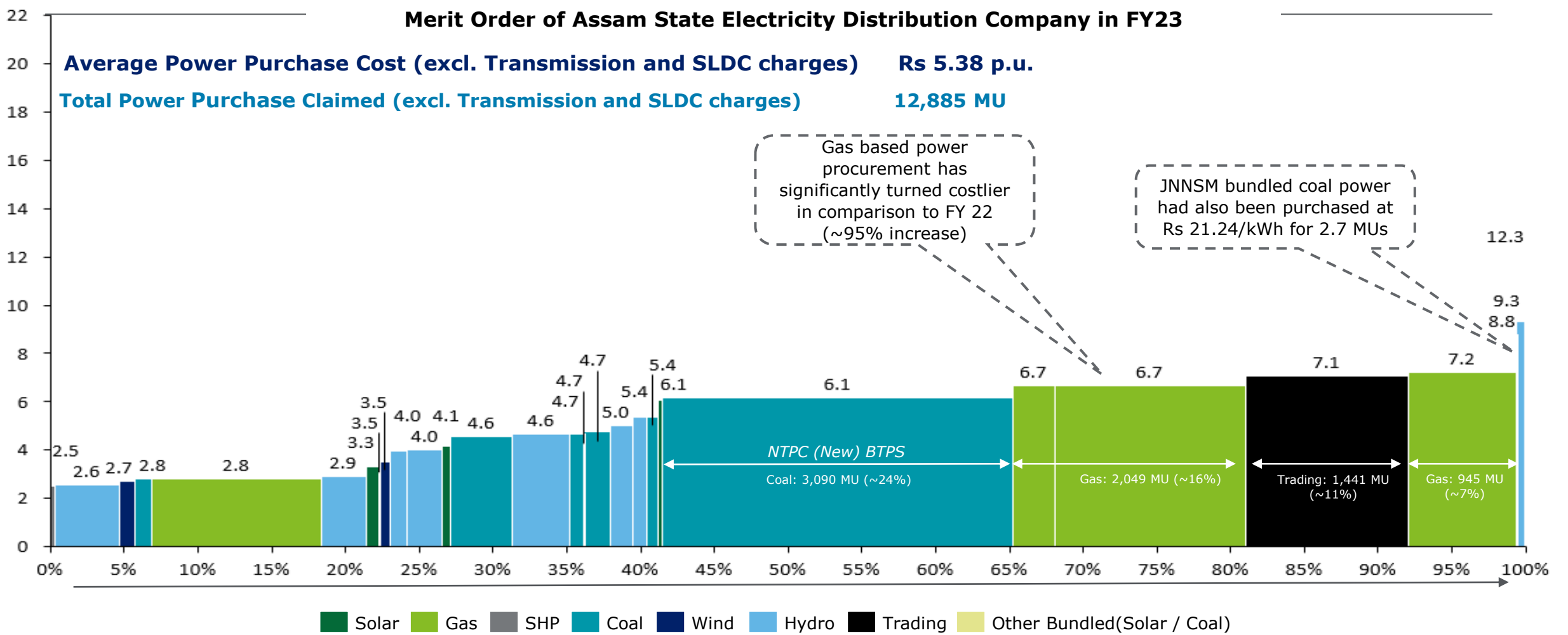
APDCL Merit Order - FY 22

About 37% of the total power supply portfolio is procured at a higher rate than average power purchase cost by APDCL; NTPC BTPS (Coal Based) with ~23% portfolio share for Assam is procured at average cost of Rs. 5.8 per unit



APDCL Merit Order - FY 2023

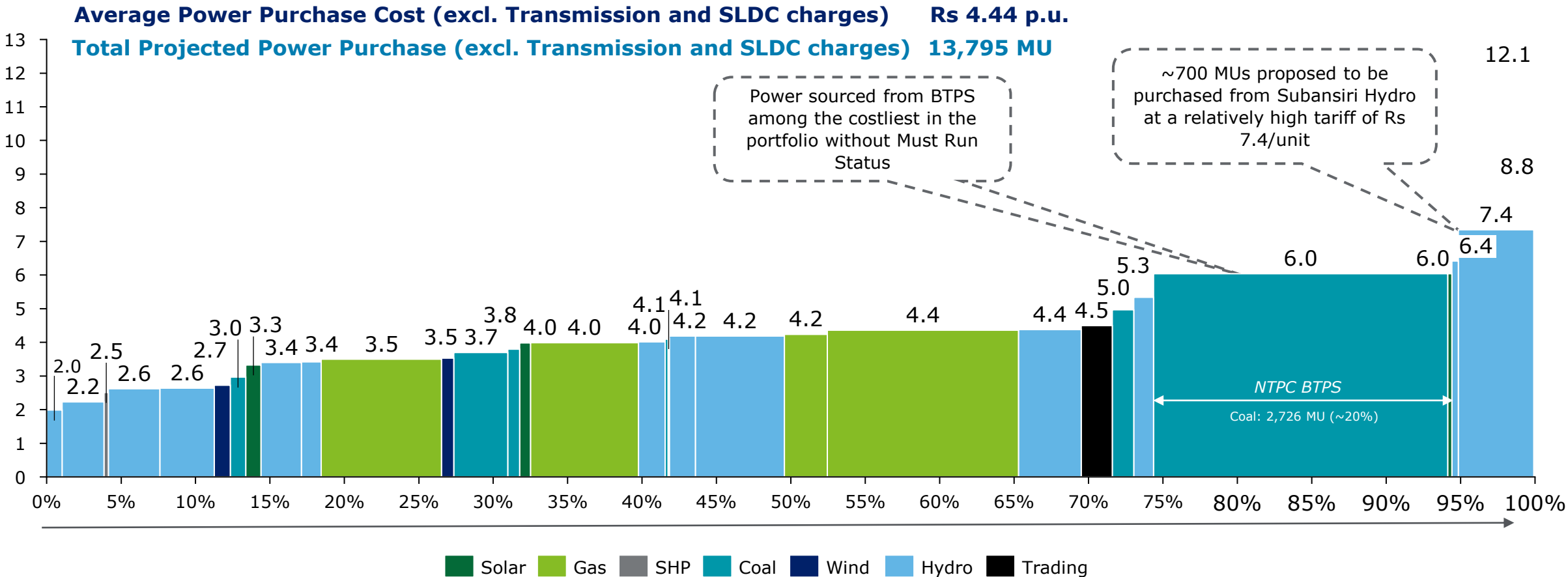
In FY 23, about 61% of the total power supply portfolio has been procured at a higher rate than average power purchase cost by APDCL; High-cost gas-based procurement (Rs. 7.2 per unit) have impacted Average power purchase cost inter alia



APDCL Merit Order - FY 2024

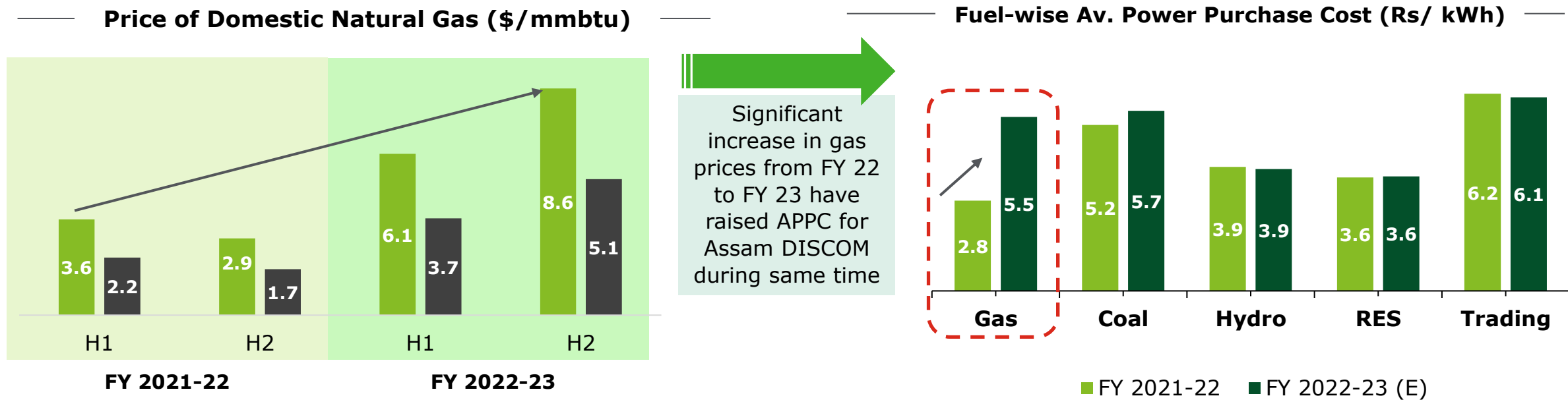
In FY 24, overall power APPC is on the lower side; owing to influx of more hydro in the portfolio; however, projected cost did not factor increase in gas prices and hence may be subjected to significant change; BTPS remains the costliest thermal option for APDCL

Merit Order of Assam State Electricity Distribution Company in FY24 (Projected)



Fuel-wise Average Power Purchase Cost for APDCL

Power sourcing from Coal (centrally owned), Gas based projects as well as short term solutions like Power Exchange contribute the most in the overall Power Purchase cost for APDCL; Further significant increase of natural gas prices from FY 22 to FY 23 has in particular severely impacted APPC in FY 23

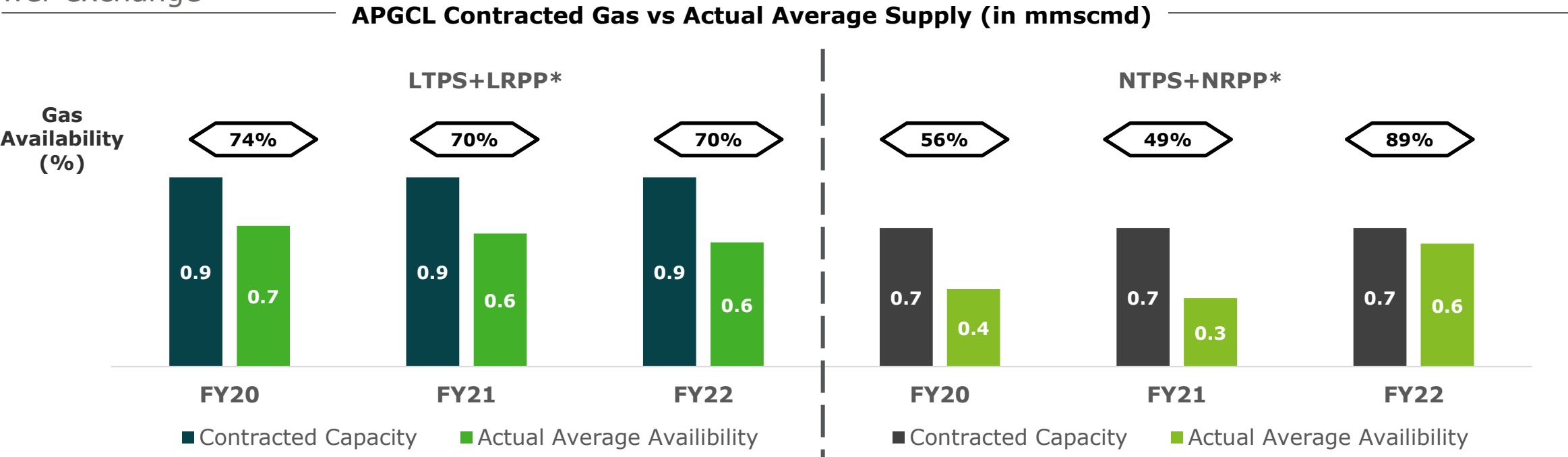


■ Domestic Natural Gas Price ■ Applicable Domestic Gas price in North-east

- ❑ During FY 11 - FY 23, period, discount of 40% for North-East customers w.r.t gas prices on NCV basis (Difference paid through Govt. budget)
- ❑ For FY 24 onwards, Govt. has approved floor price of \$4/mmbtu and ceiling price of \$6.5/mmbtu for the next two years
- ❑ Majority of power procurement are from either Centrally-owned assets or through Bilateral / Trading/ Exchange based sources as has been highlighted subsequently

Gas Availability Issues

Historically, issues of partial availability of gas have greatly impacted the power purchase portfolio of APDCL over the years wherein there has been resultant greater reliance on short term and high-cost solutions like Power exchange



- Availability of Natural gas in the North-East region are impacted owing to following reasons:**
- ❑ The Urja Ganga Pipeline is yet to be connected with North East States so these states are currently only consuming gas which they can produce.
 - ❑ There has not been any significant gas discoveries / production enhancement in the region so supply is constrained.
 - ❑ There are gas pressure issues due to field ageing and depletion challenges. Lower Gas pressure increases auxiliary consumptions and requires gas compressors which are costly for APGCL’s old power plants.

***LTPS**: Lakwa Thermal Power Station (97.2 MW); **LRPP**: Lakwa Replacement Power Project (69.755 MW); **NTPS**: Namrup Thermal Power Station (64.5 MW); **NRPP**: Namrup Replacement Power Project (98.4 MW)

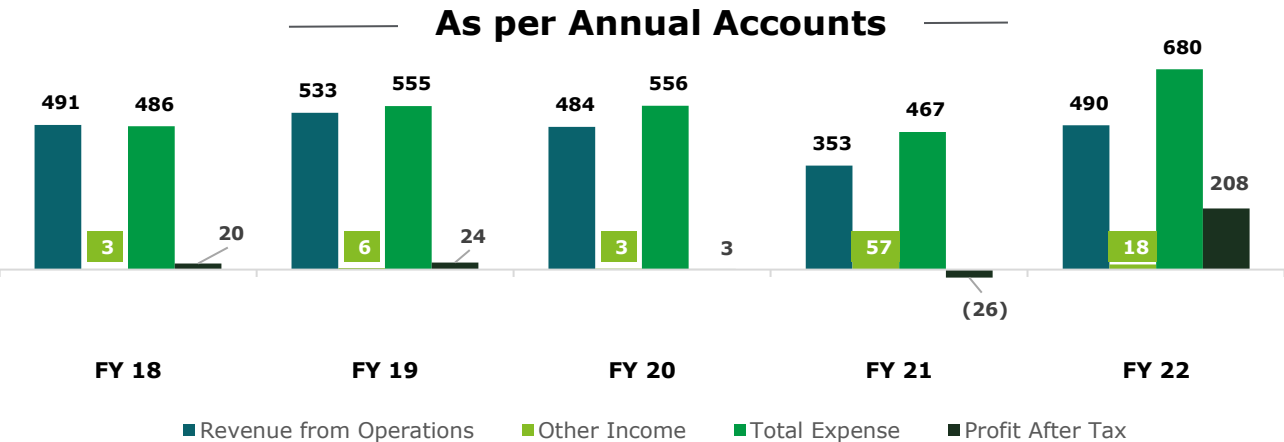
2 Assam Power Sector – Financial Snapshot



Financial Snapshot

Generation Company (APGCL)

Item	FY2018	FY2019	FY2020	FY2021	FY2022	FY 2023 (E)	FY 2024 (E)
Commercial Indicators							
Electricity transmitted (GWh)	1,403.92	1,509.77	1,505.58	1,329.75	1,996.42	2,118.92	2,199.95
Average revenue per unit (₹/kWh)	3.65	3.89	3.72	3.31	2.68	6.86	4.16
Average cost per unit (₹/kWh)	2.66	2.81	2.79	2.46	2.84	5.78	6.23
Financial Indicators (₹ Crores)							
Total Revenue	512.20	586.85	559.36	440.49	535.09	1,452.61	914.49
Expenses (excl. Depreciation and Finance Cost)	373.10	423.88	420.17	326.69	567.69	1,224.28	1,369.92
Operating profit	139.10	162.97	139.19	113.80	(32.60)	228.33	(455.43)
Interest expenses	72.85	78.77	79.85	83.89	27.29	118.41	119.54
Depreciation and amortization	40.30	52.64	56.23	56.29	85.26	94.12	98.03
Exceptional Items	-	-	-	-	396.48	-	-
Gross profit	25.95	31.56	3.12	(26.38)	251.33	15.80	(673.00)
Tax	5.54	7.27	0.54	-	43.49	2.73	-
Net Profit After Tax	20.41	24.29	2.58	(26.38)	207.84	13.07	(673.00)
Profit Ratios							
Operating profit margin (%)	27.16	27.77	24.88	25.84	(6.09)	15.72	(49.80)
Net profit margin (%)	3.99	4.14	0.46	(5.99)	38.84	0.90	(73.59)

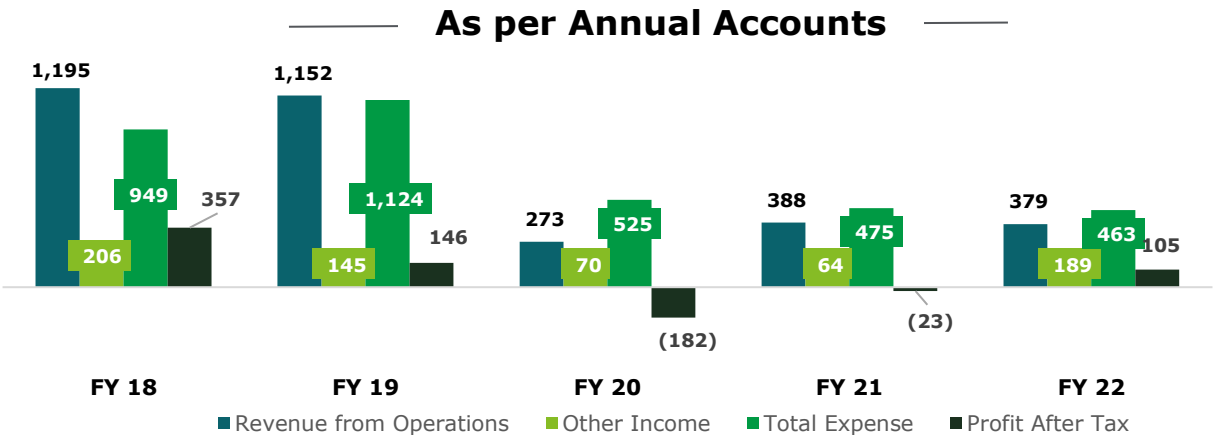


- ❑ In FY 22, increase in PAT despite higher expense due to influx of Exceptional item of amt. Rs 396.48 Cr, which in turn is due to Govt. of Assam's waiver of interest to APGCL thus becoming income of APGCL
- ❑ With respect to projected figures of expense and revenue in FY 23 and FY 24, same is taken from claimed and approved figures respectively; gap may further decrease in subsequent True-Ups of the respective years

Financial Snapshot

Transmission Company (APECL)

Item	FY2018	FY2019	FY2020	FY2021	FY2022	FY 2023 (E)	FY 2024 (E)
Commercial Indicators							
Total Energy Sent Out (GWh)	8,604.9	8,929.7	9,032.7	9,482.4	10,528.6	11,526.6	11,643.7
Average revenue per unit (₹/kWh)	1.63	1.45	0.38	0.48	0.54	0.54	0.64
Average cost per unit (₹/kWh)	0.99	1.13	0.40	0.30	0.32	0.41	0.42
Financial Indicators (₹ Crores)							
Total Revenue	1,401	1,296.5	342.9	452.0	568.3	622.4	741.3
Expenses (excluding Depreciation and Finance Cost)	848	1,009	365	286	337	473.6	491.1
Operating profit	553	288	(23)	166	232	149	250
Interest expenses	40.8	50.0	57.5	72.9	2.1	32.2	24.0
Depreciation and amortization	59.5	64.8	101.5	115.5	124.1	113.8	124.2
Exceptional Items							
Gross profit	452.2	172.8	(182)	(22.6)	105.5	2.8	102
Tax	94.8	26.9	0.0	0.0	0.0	0.4	15.9
Net Profit After Tax	357.4	145.9	(182)	(22.6)	105.5	2.4	86
Profit Ratios							
Operating profit margin (%)	39.45	22.18	(6.57)	36.68	40.77	23.90	33.76
Net profit margin (%)	25.51	11.26	(52.95)	(5.00)	18.56	0.39	11.63

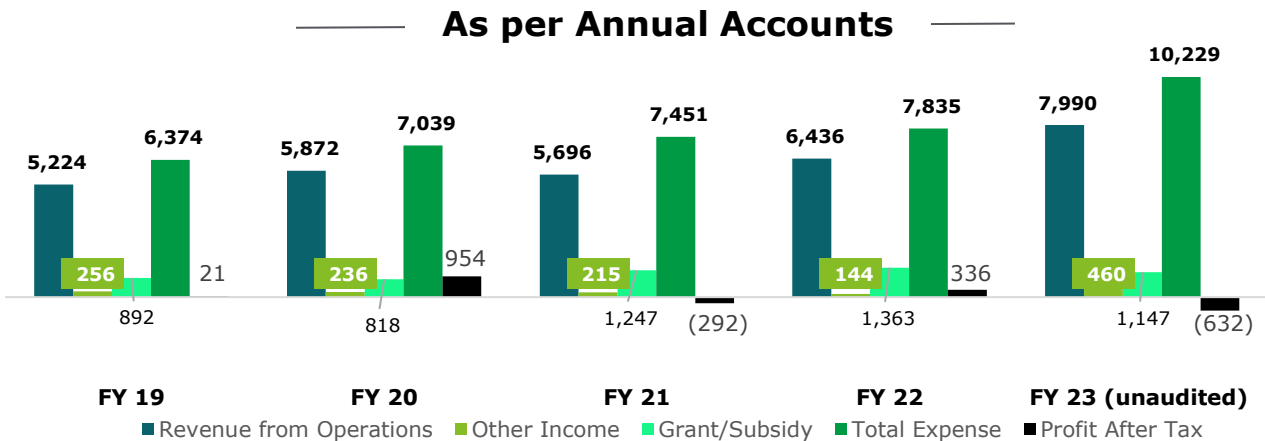


- Majorly profitable; barring FY 20 and FY 21, wherein revenue had significantly decreased compared to FY 19
- Higher expenses in FY 18 and FY 19 attributable to significantly high Transmission charges paid to PGCIL as part of Other Expenses; while significantly higher revenue from Wheeling charges were earned in the same period
- Projected figures of expense and revenue in FY 23 and FY 24, taken from APR and ARR Orders respectively

Financial Snapshot

Distribution Company - APDCL

Item	FY2018	FY2019	FY2020	FY2021	FY2022	FY 2023 (E)	FY 2024 (E)
Commercial Indicators							
Total Energy Sales (GWh)	6,814	6,968	7,257	7,458	8,520	9,043	9,765
Average Cost of Supply (₹/kWh)	7.85	8.72	8.97	9.36	8.73	10.87	10.1
Average Billing Rate (₹/kWh)	6.97	7.50	8.09	7.64	7.55	8.84	8.369
Financial Indicators (₹ Crores)							
Total Revenue	5,899	6,372.0	6,926.3	7,158.1	7,943.5	9,596.9	8,172.7
Expenses (excluding Depreciation and Finance Cost)	5,426	6,009	6,650	6,956	7,278	9,607	9,705
Operating profit	473	363	277	202	666	(10)	(1,532)
Interest expenses	267.4	290.7	168.7	203.3	158.9	144.1	34.2
Depreciation and amortization	45.8	74.6	220.8	291.3	398.1	478.4	143.6
Exceptional Items	4.9	23.0	1066.6	-	227.8	-	-
Gross profit	164.4	21.0	953.8	(292.4)	336.4	(632.2)	(1,710)
Tax	-	-	-	-	-	-	-
Net Profit After Tax	164.4	21.0	953.8	(292.4)	336.4	(632.2)	(1,710)
Profit Ratios							
Operating profit margin (%)	8.01	5.70	4.00	2.82	8.38	(0.1)	(18.7)
Net profit margin (%)	2.79	0.33	13.77	-4.09	4.23	(6.6)	(20.9)



- Revenue steadily increased till FY 23 at a CAGR of ~8.5%, but expenses grew at a faster rate of ~10.1% (CAGR),
- Owing to influx of Exceptional items of value Rs ~228 Cr (reversal of interest on GoA loans), PAT was positive in FY 22
- In FY 23, significantly higher power purchase cost have contributed to Loss, this gap may get adjusted based on approval of the Commission

3. RE Policy and Regulatory Framework



Policy and Regulatory Framework

National Level

Policies

National Electricity Policy, 2005

Provides Guidelines for:

- accelerated development of the power sector,
- providing supply of electricity to all areas and protecting interests of consumers
- Energy security

National Tariff Policy, 2016

- SERCs encouraged to allow 35% (of installed capacity) power procurement from state projects u/s 62 of EA 2003
- Lays down framework for performance-based cost of service regulation in respect of aspects common to generation, transmission and distribution

Hydro Policy, 2019

Salient Features:

- Declaration of Large Hydro (>25 MW) as RES
- Setting Hydro Purchase Obligation (HPO)
- Tariff rationalization measures
- Provision for budgetary support w.r.t flood responses

Green Hydrogen Policy, 2022

Green Hydrogen Policy provides the details of production of Green Hydrogen using Renewable Energy and compliance and implementations of all stake holders

Schemes/ Regulations/Rules

MoP Waiver of ISTS Charges, 2021

- Waiver on ISTS charges applicable to Solar, Wind, Pumped Storage Plants and Battery energy Storage projects
- Extension of the waiver of ISTS charges on transmission of electricity generated from Solar & Wind projects commissioned upto June, 2025

Solar Parks and Ultra Mega Solar Projects Scheme, 2014

- Provisions for promotion of solar park and ultra mega RE power parks
- Revised target capacity of 40 GW (initially 20 GW)
- Timeline of scheme till 31st March 2026
- CFA assistance to SPPDs upto Rs 20 Lakhs/MW or 30% project cost whichever is lesser

RE Bundling Scheme, 2022

- Mechanism for bundling of cheaper RE with costlier thermal power
- Discoms have flexibility to procure RE power within existing PPA
- Gains through bundling to be shared between the generators and the beneficiaries
- Allows JVs to set up RE plants in generating plants

MNRE Rooftop Subsidy Scheme (Ph II), 2019

- Target to achieve 40 GW under Phase II of this scheme with validity till end of FY 26
- Two Components:
 - ❑ Comp. A –Residential Sector
 - ❑ Comp. B – Incentives to DISCOMs for initial 18 GW
- Provisions for category-wise CFA assistance

Policy and Regulatory Framework

State Level

State Level Policies

Small Hydropower (SHP) Development Policy, 2007

- Applicable to all hydro projects upto 25 MW capacity
- Process of allotment of project defined
- Waiver of wheeling charges for power sold to State DISCOM
- 6-months banking facility allowed
- Exemption of Entry Tax
- Provision for free transfer of shares
- Time limit for project execution from date of receipt of all statutory approvals :
 - FC – 12 months
 - SCOD – 36 months

Installed capacity of Small hydro plants (capacity upto 25 MW) has increased from 2 MW in 2007 to 42.55 MW in Assam (~2000% increase)

Assam Renewable Energy Policy 2022

- Target of 1200 MW of additional RE power capacity by 2027
- Provision for setting up of 1000 MW solar plants on free unused government land As per “Mukhya Mantri Souro Shakti Prokolpo” -.
- Solar park development promoted; APDCL to act as SPPD
- 250 MW assured offtake by APDCL for solar projects set up to supply solely to APDCL
- Direction given for development of necessary T&D infrastructure to align with future RE capacity addition

~24% increase in RE capacity addition from FY 22 to FY 23 as per CEA

State Level Schemes/ Regulations

AERC Scheduling, Forecasting, DSM Regulations 2018

- Facilitation of grid interaction of solar and wind energy generated in Assam while maintaining grid stability and security envisaged under the Assam Grid Code and the Act through forecasting, scheduling, and a mechanism for the settlement of deviations if any

AERC (Grid Interactive Solar PV Systems) Regulations, 2019

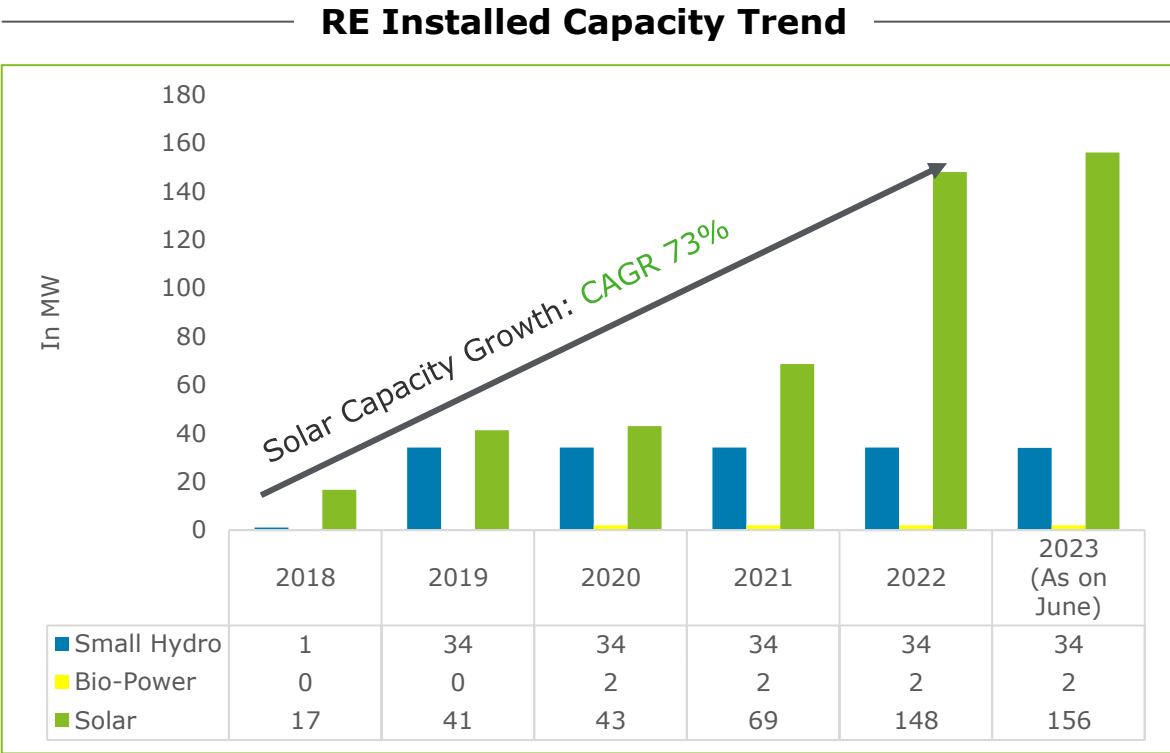
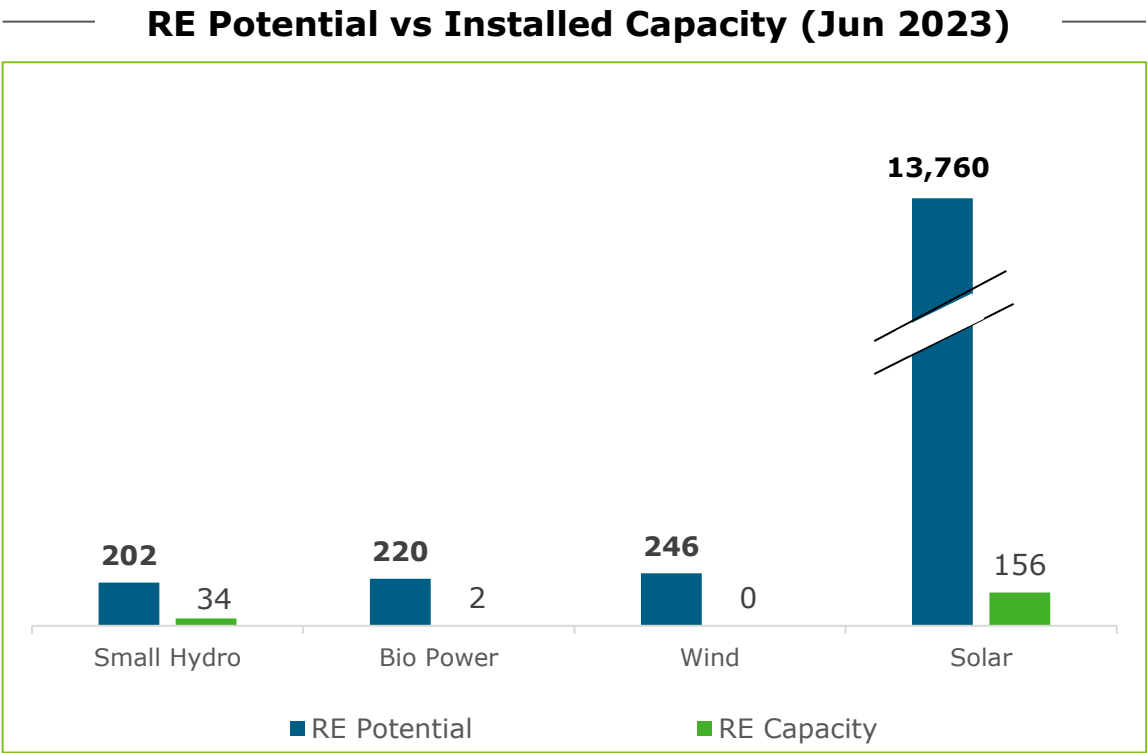
- Capacity Targets for Distribution Licensees
- Eligible consumers, individual project capacity and interconnection voltage
- Interconnection standards and safety requirements
- Energy accounting and settlement; Metering arrangement

AERC RPO amended Regulations, 2021

- Inclusion of Large Hydro and Pumped Storage projects as RES for projects commissioned after 8th Mar 2019
- Inclusion of energy generated from Waste Heat Recovery System as RE source for fulfilment of RPO
- Inclusion of solar projects with storage for fulfilment of solar RPO
- Option for HPO to be met by Hydro Energy Certificates (HEC)
- Conditional RPO (based on time of commissioning) for CPPs

Renewable Energy in Assam - Potential and Capacity

Major opportunity for proliferation of RE in Assam is provided by the significant solar potential of ~14 GW out of which only 0.15 GW has been achieved (~1%) so far



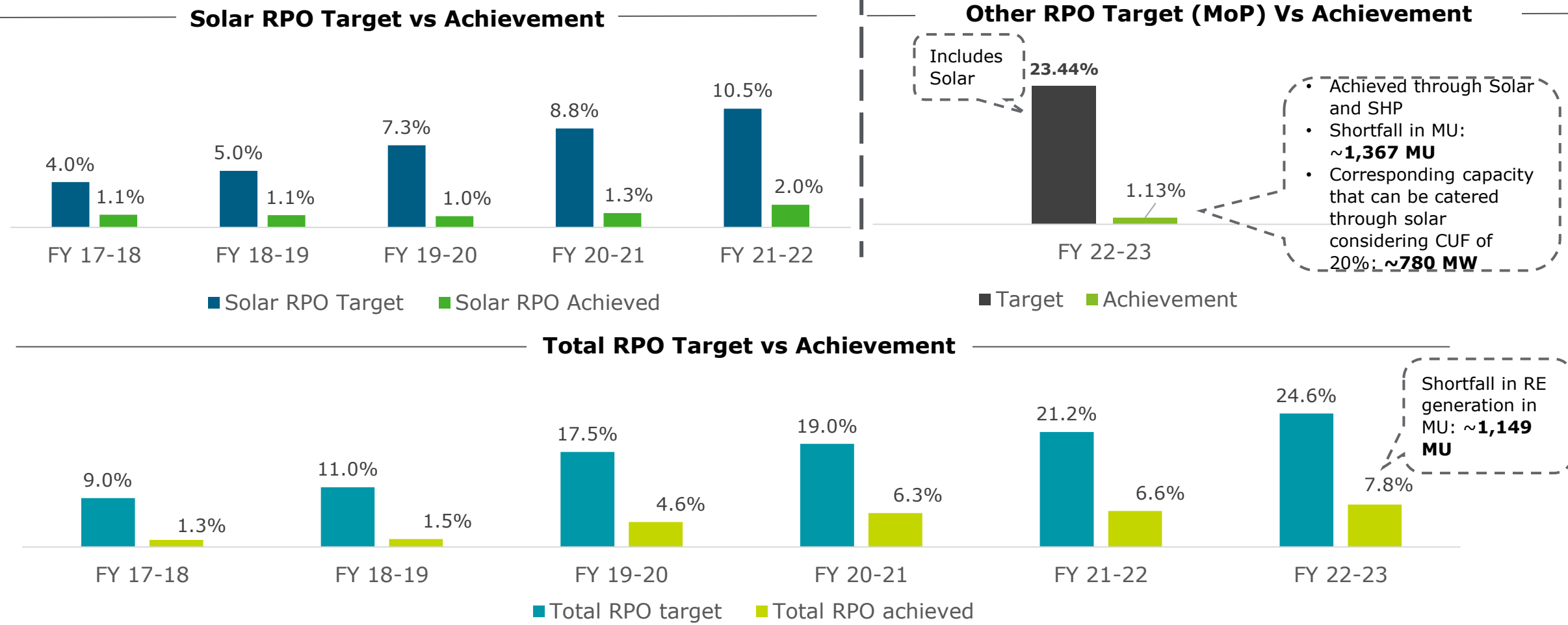
Note: RE installed capacity data in Assam is as per MNRE. Breakup available for 241.5 MW as per NERLDC and CEA; APDCL may confirm actual figure.

- ❑ Major capacity addition of RE in state has been through the Private Sector
- ❑ Significant scope of RE capacity addition through solar in state with respect to potential of ~14 GW in state

RPO Target vs Achievement in Assam

(1)

Increasing trend of shortfall w.r.t RPO achievement in Assam against targets as the years have progressed; significant scope for further RE proliferation



Highest RE potential in state is that of solar (~14 GW); giving **opportunity to cater to the upcoming high RPO targets**

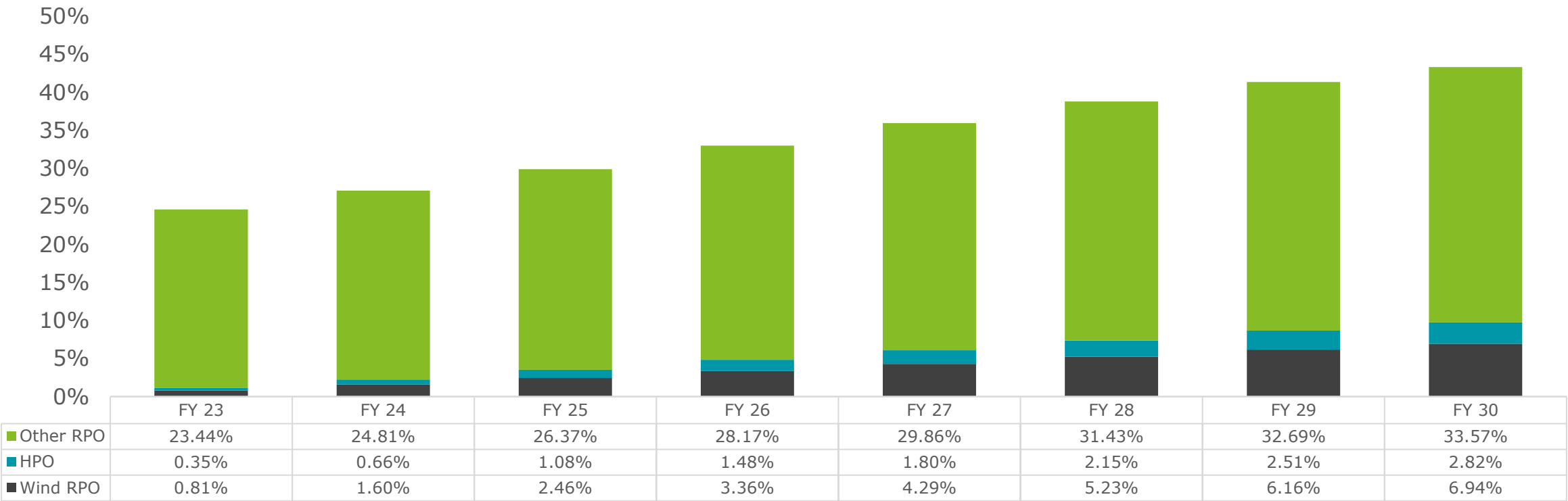
Note: This is based on Power Purchase Portfolio as per True-Up Orders of APDCL (FY 17-18 to FY 21-22) and APR order for FY 22-23. Any procurement of power from eligible large hydro projects (commissioned after 28th Mar 2019) and REC mechanism has not been considered above. Hence, above data is susceptible to updation once such data, if any is made available by APDCL

RPO Target vs Achievement in Assam

(2/2)

Targets have increased significantly from FY 22-23 onwards as per MoP Order

Revised RPO Targets from FY 23 onwards as per MoP Order



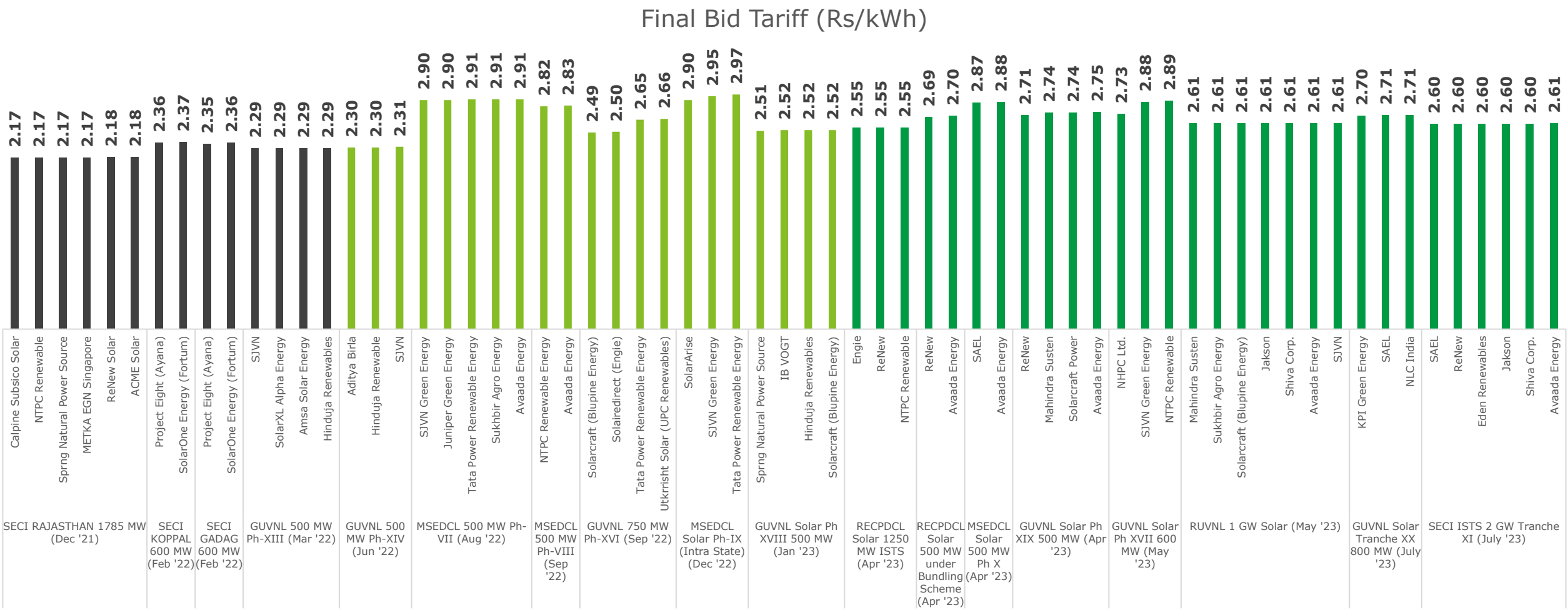
Note: Other RPO includes all RE sources except Wind and Hydro i.e., **Solar**, hydro (other than LHP after 08.03.19), Biomass etc

- Assam Electricity Regulatory Commission (Renewable Purchase Obligation and Compliance) Regulations, 2022 has aligned the trajectory with the Ministry of Power Order No. 23/03/2016-R&R dated 29.01.2021.
- With respect to revised trajectory, from FY 23 onwards, solar procurement/ generation can be utilized to fulfil Other RPO targets

4. Solar Sector In India



Solar Tariff Trends



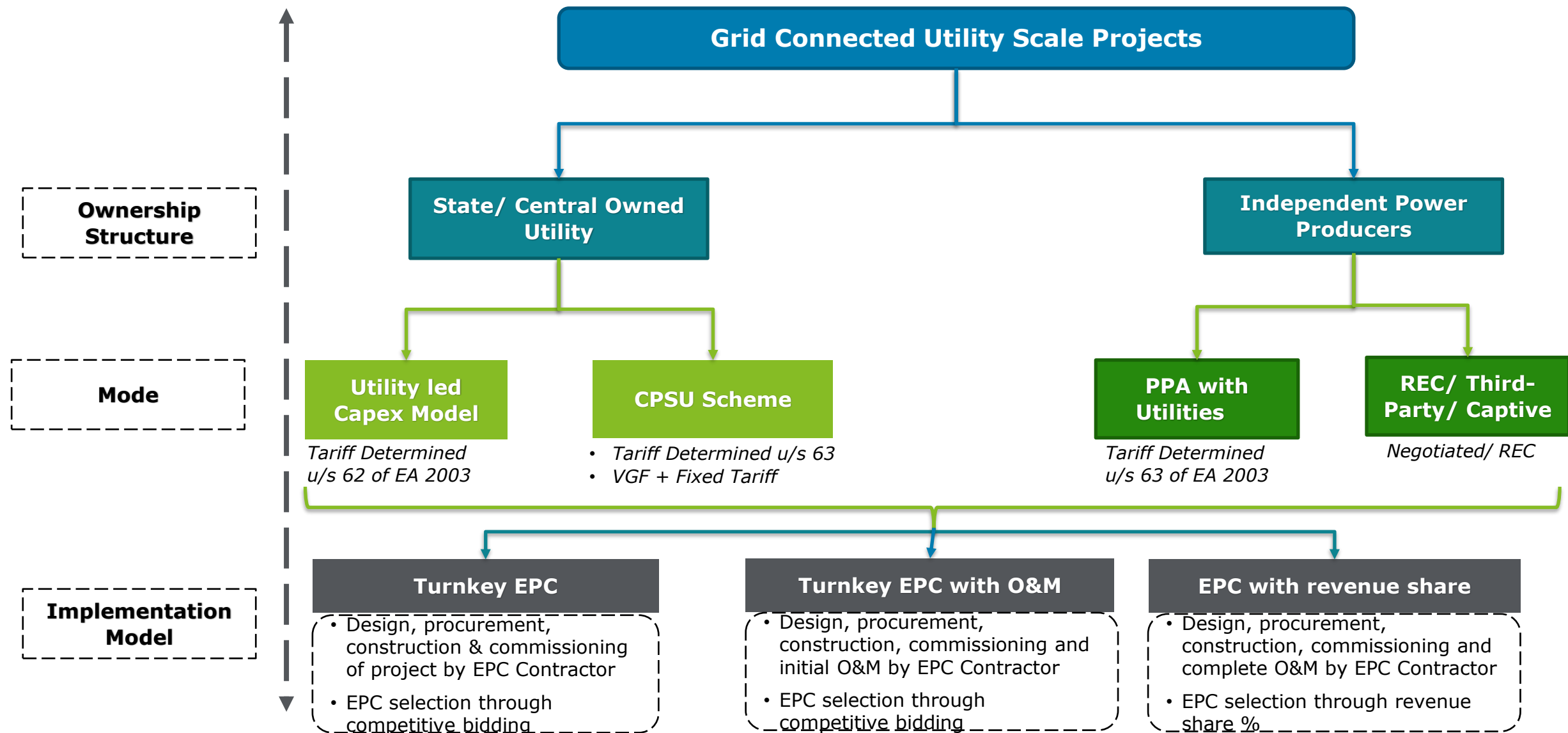
❑ Imposition of Basic Customs Duty and ALMM mandate have resulted in increase of solar tariffs from FY 23 onwards

❑ However, decrease in module prices inter alia have encouraged increased quantum of solar auctions carried out in FY 24 as compared to FY 23

❑ Current Global module prices have dropped to USD 0.18/Wp while Domestic module prices have reached Rs 24.7/Wp for 440 Wp and Rs 26.3/Wp for 500 Wp modules incl GST and freight

Business Models – Grid Connected Utility Scale Solar Projects

Models have been showcased basis type of ownership



Utility led Capex Model

NTPC 92 MW Kayalkulam Floating Solar Plant, Kerala

- Memorandum of Understanding (MOU) signed between the NTPC Ltd. and the Kerala State Electricity Board (KSEB) in May 2018.
- NTPC selected the EPC contractor through a competitive bidding process for setting up 92 MW floating solar plants in two phases at Rajiv Gandhi Combined Cycle Power Project (RGCCP), Kayamkulam, Kerala

Key Features

Capacity

- Phase I: 22 MW (EPC Contractor: M/S BHEL)
- Phase I: 70 MW (EPC Contractor: M/S Tata Power Solar Systems Limited)

Power Offtake

- PPA with KSEB at Rs 2.94/unit (Tariff determined u/s 62 of EA 2003)

Capital Cost (including IDC) as approved by CERC

- Phase I (22 MW): ~Rs 115 Cr
- Phase II (70 MW): ~Rs 404 Cr

Funding

- Balance Sheet Financing by NTPC
- D/E ratio of 80:20 as considered by Commission

- EPC selection through competitive bidding adopted by Regulator for tariff determination
- Plant fully operational from June 2022

CPSU Scheme

~7 GW already awarded under this scheme

- Under CPSU scheme Phase I, Government/ Government entities could participate in tenders wherein tariff was capped, and basis of bid participation would be quantum of Viability Gap Funding (VGF) quoted
- Responsibility of finding buyers would be with the Bid winners
- Government of India has approved the implementation of CPSU Scheme Phase-II for setting up 12,000 MW grid-connected Solar Photovoltaic (PV) Power Projects by CPSUs/ State PSUs/ Government Organizations, with Viability Gap Funding (VGF) support for self-use or use by Government/ Government entities, either directly or through Distribution Companies (DISCOMS) Phase

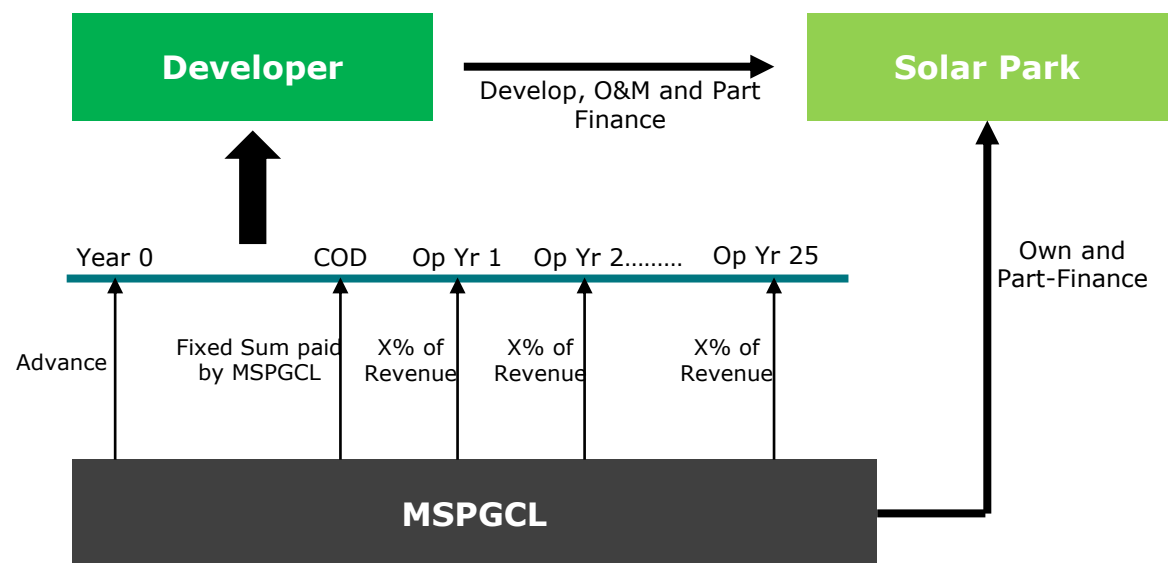
CPSU Tender	Date	Tendered Capacity (MW)	Awarded Capacity (MW)	Capped Tariff	VGF Range (Rs Lakh)
SECI Tranche I	Aug 2019	2,000	922	Rs. 3.50/kWh	55.00-70.00
SECI Tranche II	Nov 2019	1,500	1,104	Rs. 3.50/kWh	68.00-70.00
IREDA Tranche III	Sep 2021	5,000	5,000	Rs. 2.45/kWh	44.72-44.95
CAPACITY (MW)		8,500	7,026		

• In SECI Tranche I tender, APDCL has also been awarded with 30 MW capacity under this CPSU scheme

PPP based Performance-linked Revenue Sharing Model

50 MW project executed by Welspun for MSPGCL under revenue sharing model

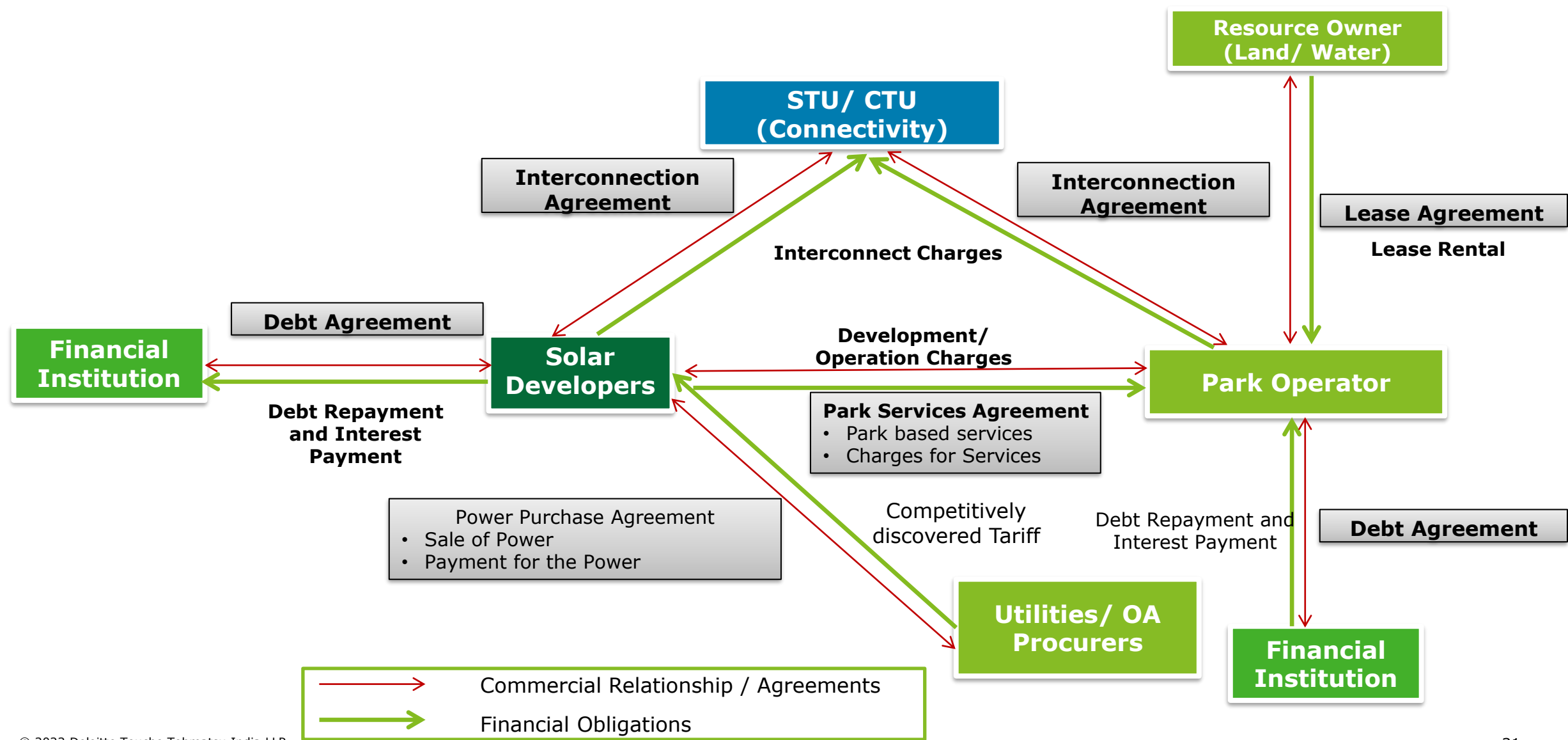
- This model is based on the concept of sharing of risk and return of the project with the project developer over the project life.
- The contract covers two periods, one the construction period which starts from signing of documents till COD of project and other is O&M period which starts from COD till expiry of 25 years.
- Apart from undertaking the EPC and the O&M for the project, the developer has to part finance the capital investment. In return, Developer is entitled to receive a performance based revenue share from this project



Key Aspects	
Scope of work for MSPGCL	- Signing of PPA with MSEDCL - Land, Clearances, Connectivity
Scope of work for Project Developer	- Design, engineering, manufacture, supply, erection, testing and commissioning of the project - O&M for 25 years
Mode of selection of Developer	The successful bidder was selected based on the net present value of the highest revenue share offered to MSPGCL over the life of the project.
Financing	- Contribution of Rs 3 Cr/MW by MSPGCL - Balance financing by Developer (Welspun)
Performance Requirement	- Maintenance of at least 19% CUF - Commissioning of project within 9 months
Payment Security	- Revenue from the project routed through a trust and retention account to protect the interest of either party - Revolving LC (3 months billing equivalent) to Developer

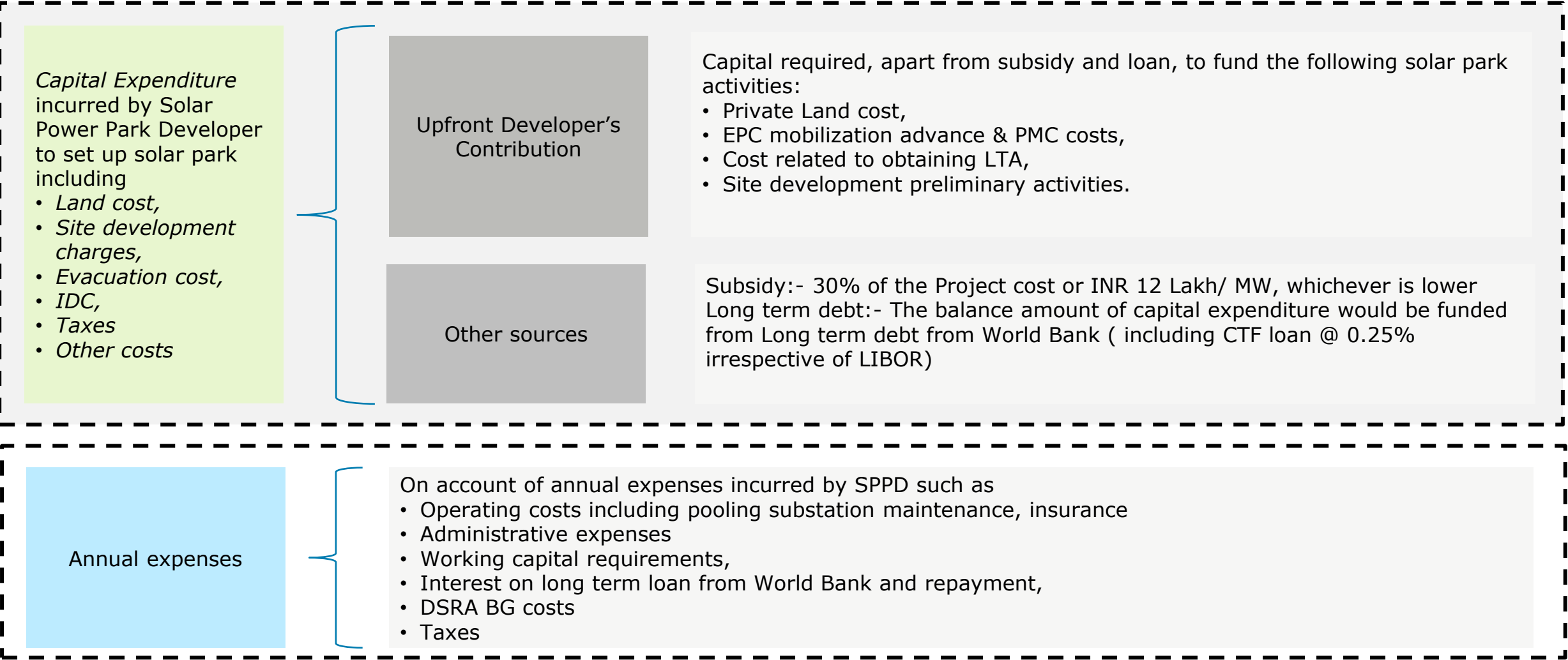
PPP based solar park model

Commercial Arrangements



PPP based solar park model

Determination of Charges



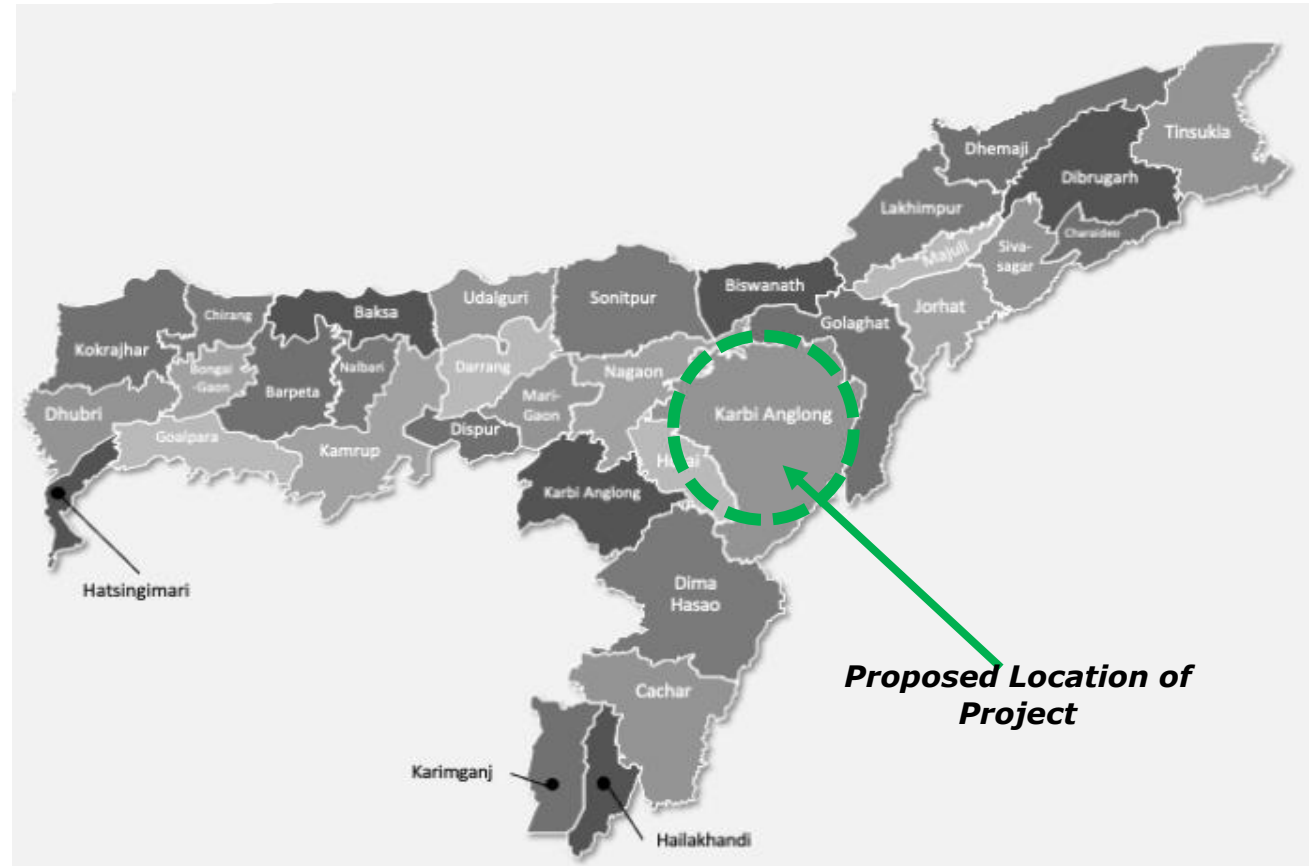
5 Solar Project



1000 MW APDCL Solar Project – Phase I

Project currently under DPR preparation stage

- ❑ 1000 MW envisaged to be developed by APDCL on Utility led Capex Model in Phase I
- ❑ 70:30 Debt Equity Ratio
 - Debt: 90% of Debt as grant from Govt. of India remaining from ADB
 - Equity: To be provided by APDCL/Govt. of Assam
- ❑ Land has already been identified by Govt. of Assam and DPR preparation is currently underway
- ❑ Total ADB Lending will be \$435 Million



Thank You

State-wise Details Of Solar Parks (1/3)

~39.28 GW approval for solar parks have been accorded out of which, ~8 GW solar parks have been completed

#	State	Name of Solar Park	Approved Capacity (MW)	Status
1	Andhra Pradesh	Ananthapuramu-I Solar Park	1400	Completed
2		Kurnool Solar Park	1000	Completed
3		Kadapa Solar Park	1000	
4		Ananthapuramu-II Solar Park	500	Completed
5		Solar-Wind Hybrid Solar Park	200	
6	Chhattisgarh	Rajnandgaon Solar Park	100	
7	Gujarat	Radhanesda Solar Park	700	
8		Dholera Ph-I Solar Park	1000	
9		NTPC RE Park	4750	
10		GSECL RE Park	3325	
11		GIPCL RE Park Ph-I	600	
12		GIPCL RE Park-Ph-II	1200	
13		GIPCL RE Park-Ph-III	450	
14	Himachal Pradesh	Kaza Solar Park	880	
15		Kinnaur Solar Park	400	
16	Jharkhand	Floating Solar Park	100	
17		Deogarh Solar Park	20	
18		Palamu Solar Park	20	
19		Garwha Solar Park	20	
20		Simdega Solar Park	20	
21	Jharkhand & West Bengal	DVC Floating Solar Park Ph-I	755	
22		DVC Floating Solar Park Ph-II	234	
23	Karnataka	Pavagada Solar Park	2000	Completed
24		Kalburgi Solar Park	500	
25	Kerala	Kasargod Solar Park	105	
26		Floating Solar Park	50	

State-wise Details Of Solar Parks (2/3)

~39.28 GW approval for solar parks have been accorded out of which, ~8 GW solar parks have been completed

#	State	Name of Solar Park	Approved Capacity (MW)	Status
27	Madhya Pradesh	Rewa Solar Park	750	Completed
28		Mandsaur Solar Park	250	Completed
29		Neemuch Solar Park	500	
30		Agar Solar Park	550	
31		Shajapur Solar Park	450	
32		Chhatarpur Solar Park	950	
33		Omkareshwar Floating Solar Park	600	
34		Morena Solar Park	1400	
35		Barethi Solar Park	630	
36	Maharashtra	Sai Guru Solar Park	500	
37		Dondaicha Solar Park	250	
38	Mizoram	Vankal Solar Park	20	
39	Odisha	Solar park by NHPC	40	
40		Floating Solar Park Ph-I	100	
41		Floating Solar Park Ph-II	200	
42	Rajasthan	Bhadla Phase-II Solar Park	680	Completed
43		Bhadla III Solar Park	1000	Completed
44		Bhadla IV Solar Park	500	Completed
45		Phalodi-Pokaran Solar Park	750	
46		Fatehgarh Phase 1B Solar Park	421	
47		Nokh Solar Park	925	
48		Pugal Solar Park Phase-I	1000	
49		Pugal Solar Park Phase-II	450	
50		RVUN Solar Park	1310	

State-wise Details Of Solar Parks (3/3)

~39.28 GW approval for solar parks have been accorded out of which, ~8 GW solar parks have been completed

#	State	Name of Solar Park	Approved Capacity (MW)	Status
51	Uttar Pradesh	Solar Park in UP	365	
52		Jalaun Solar Park	1200	
53		Kalpi Solar Park	65	
54		Mirzapur Solar Park	100	
55		Lalitpur Solar Park	600	
56		Jhansi Solar Park	600	
57		Chitrakoot Solar Park	800	
			39285	

List of commissioned SECI projects under BOO model (1/3)

~5.6 GW projects commissioned as on date*

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
2000 MW ISTS Connected Solar PV Projects (ISTS-I)					
1	Mahindra Renewables Private Limited	250	Vill-Bhiv Ji ka Gaon, Tehsil-Bap, Dist.-Jodhpur	14.05.2021 (100 MW) 07.06.2021 (60 MW) 23.07.2021 (50 MW) 17.08.2021 (40 MW)	250
2	Kilaj Solar (Maharashtra) Pvt. Ltd	50	Vill:, Nure Ki Bhurj/Sihra, Dist: Jodhpur	17.Apr.2020	50
5	Azure Power Forty Three Private Limited	300	Village Jagdev wala and Daudsar, Tehsil Bikaner, District Bikaner	15.12.2020 (150 MW) 30.01.2021 (100 MW) 08.02.2021 (50 MW)	300
6	Azure Power Forty Three Private Limited	300	Village Jagdev wala and Daudsar, Tehsil Bikaner, District Bikaner	25.06.2021 (50 MW) 08.08.2021 (50 MW) 22.09.2021 (100 MW) 31.12.2021 (100 MW)	300
7	Clean Solar Power (Jodhpur) Pvt. Ltd	250	Vill-Noore Bhurj, Tehsil-Bap-Dist.-Jodhpur	14.03.2022 (115 MW) 21.04.2022 (135 MW)	250
750 MW Grid Connected in Rajasthan Tranche-I (non-solar park)					
2	Fortum Solar Plus Pvt. Ltd.	250	Vill: Bainsada , Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.03.2021 (150 MW) 01.05.2021 (100 MW)	250
3	ReNew Solar Energy (Jharkhand Five) Pvt. Ltd	110	Vill: Bhainsara, Tehsil: Fatehgarh, Dist: Jaisalmer, State Rajasthan	04.02.2021 (50 MW) 11.03.2021 (60 MW)	110
4	Sitara Solar Energy Pvt. Ltd.	100	Vill: Kanasar, Tehsil: Bap, Dist: Jodhpur, State Rajasthan	20.May.2021	100
5	Palimarwar Solar House Pvt. Ltd.	20	Vill: Hariyasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.04.2021 (20 MW)	20
6	Palimarwar Solar House Pvt. Ltd.	20	Vill: Hariyasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	19.04.2021 (20 MW)	20

List of commissioned SECI projects under BOO model (2/3)

~5.6 GW projects commissioned as on date*

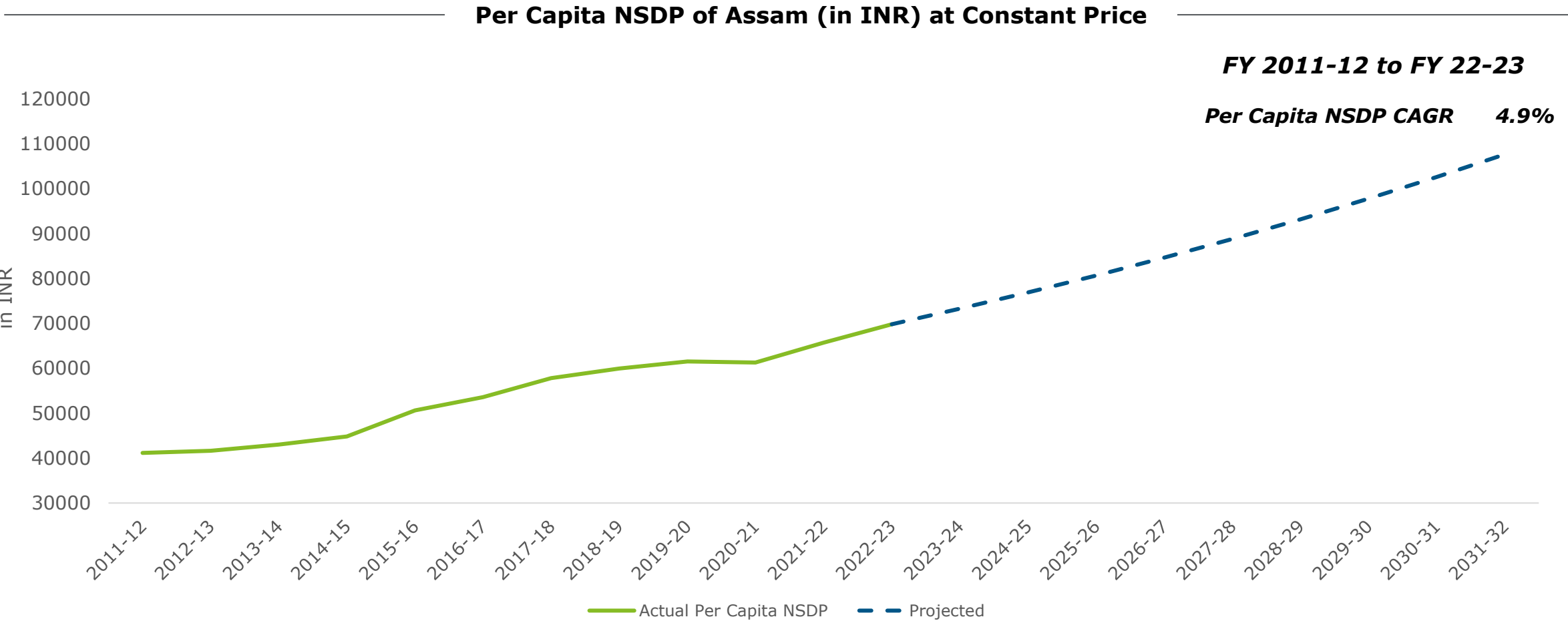
Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
1200 MW ISTS Connected Projects (ISTS-III)					
1	SBSR Power Cleantech Eleven Pvt. Ltd.	300	Vill: Lalsar & Hapasar, Tehsil: Bikaner, Dist: Bikaner, State Rajasthan	15.08.2021 (50 MW) 04.04.2022 (50 MW) 11.04.2022 (50 MW)	150
2	ReNew Sun Waves Pvt. Ltd.	300	Vill: Mandhopura, Tehsil: Fatehgarh, Dist: Jaisalmer, State Rajasthan	13.08.2021 (150 MW) 17.08.2021 (50 MW) 04.09.2021 (50 MW) 04.10.2021 (50 MW)	300
3	Eden Renewable Cite Pvt. Ltd.	300	Vill:Lakhasar, Tehsil: Pokhran, Dist: Jaisalmer, State Rajasthan	13.08.2021 (300 MW)	300
4	Azure Power Forty One Pvt. Ltd.	300	Vill: Noore Ki Bhurj, Khakhuri, Dedasari, Kushla Ram ki Basti, Ismail ki Dhani, Tehsil: Bap, Dist: Jodhpur, Rajasthan	12.10.2021 (50 MW) 30.10.2021 (50 MW) 30.11.2021 (50MW) 24.12.2021 (50 MW) 31.01.2022 (50 MW) 07.03.2022 (50 MW)	300
1200 MW ISTS Connected Projects (ISTS-IV)					
1	Ayana Renewable Power One Pvt. Ltd.	300	Vill: Khichiya, Bikaner Dist., Rajasthan	25.11.2021 (100 MW) 07.12.2021 (50 MW) 22.12.2021 (150 MW)	300
2	Renew Solar Energy (Jharkhand Three) Pvt. Ltd.	300	Vil: Mandhopura, Tehsil: Fatehgarh, Dist Jaisalmer, Rajasthan	02.09.2021 (150 MW) 27.09.2021 (50 MW) 04.10.2021 (50 MW) 09.12.2021 (50 MW)	300
3	Azure Power Maple Pvt. Ltd.	300	Vill: Sonanda, Shekhar, Bandhari & Kesarapura Teshsil: Bap, Dist. Jodhpur, Rajasthan	14.02.2022 (53 MW) 30.03.2022 (204 MW)	300
4	Mega SuryaUrja Pvt. Ltd.	250	Jodhpur Dist., Rajasthan	20.05.2022 (175 MW)	250

List of commissioned SECI projects under BOO model (3/3)

~5.6 GW projects commissioned as on date*

Sr. No.	Name of Project Developer	Project Capacity (MW)	Project Location (State)	Actual Commissioning Date	Total capacity commissioned as on date (MW)
Solar Tenders under SBG					
750 MW Grid Connected in Rajasthan Tranche-II (non-solar park)					
1	NTPC Limited	160	Vill: Jetsar, Dist. Sri Ganga Nagar, Rajasthan	22.10.2021 (80 MW) 25.03.2022 (80 MW)	160
2	Mahindra Susten Pvt. Ltd.	200	Bikaner Dist. Rajasthan	14.10.2021 (200 MW)	200
3	Clean Solar Power (BHAINSADA) Pvt. Ltd.	250	Vill: Bhainsada, Amarsar/Balwantgarh, Tehsil: Bhaniyana, Dist: Jaisalmer, Rajasthan	05.04.2022 (150 MW) 05.05.2022 (100 MW)	250
1200 MW ISTS Connected Projects (ISTS-V)					
1	GRT Jewellers (India) Pvt. Ltd.	150	Taluka: Ettayapuram, Kayathar, District: Thoothukkudi, Tamil Nadu	30.03.2022 (50 MW) 13.04.2022 (50 MW)	150
1200 MW ISTS Connected Projects (ISTS-VI)					
1	Avaada Sustainable RJProject Private Limited	300	Village: Noorsar, Tehsil: Bikaner in Bikaner, Rajasthan	14.01.2022 (100 MW) 07.02.2022 (50 MW) 12.04.2022 (50 MW) 11.05.2022 (100 MW)	300
2	Masaya Solar Energy Private Limited	300	Khandwa, Madhya Pradesh		150
3	ReNew Solar Urja Private Limited	300	Village Mandhopura, Tehsil Fatehgarh, District Jaisalmer, State Rajasthan	01.12.2021 (150 MW) 07.12.2021 (50 MW) 13.12.2021 (50 MW) 16.12.2021 (50 MW)	300
2000 MW ISTS Connected Projects (Tranche-IX)					
1	Thar Surya 1 Pvt. Ltd.	300	Vill: Dholera 1, Jamsar, Dist: Bikaner, Rajasthan	23.03.2022 (54 MW) 23.06.2022 (54 MW) 09.08.2022 (54 MW) 11.10.2022 (74 MW) 24.11.2022 (64 MW)	300

Overall growth in Assam economy as indicated by growth at ~5% CAGR w.r.t Per Capita Net State Domestic Product (NSDP) at Constant Price as per MOSPI



Details of Upcoming Projects Envisaged

Estimated timeline	Capacity(MW)	Expected CUF/PLF	Estimated Annual Generation in MUs	Name of project	Type of resource	Agency	Remarks/Source link
2023-24	492.72	65%	2665	3 X 660 MW NUPPL project	Thermal	APDCL	Allocation from project as per PPA signed in Feb 2023 https://cea.nic.in/wp-content/uploads/thermal_broad/2023/05/B_S_MAY_23_V1.1.pdf
2023-24	20	20%	35	Namrup Solar PV Project	Solar	APGCL	Delayed project. Estimated project cost of Rs 94.2 Cr
2023-24	60	20%	105	Sonabeel Floating Solar Project	Solar	APGCL	Exp commissioning by Dec 2023. Estimated project cost of Rs 99.54 Cr
2023-24	20	20%	35	Majuli Solar PV Project	Solar	APGCL	Exp commissioning by Dec 2023.
2024-25	120	45%	473	Lower Kopili HEP	Hydro	APGCL	Exp commissioning by Aug 2024. Estimated project cost of Rs 2127 Cr
2025-26	24	45%	95	Borpani Middle II SHEP	Hydro	APGCL	Exp commissioning by Nov 2025. Estimated project cost of Rs 232.53 Cr
By 2025-26	100	20%	175	100 MW APDCL floating solar project on PPP basis	Solar	APDCL	https://www.mercomindia.com/tender-issued-to-develop-100-mw-of-floating-solar-projects-in-assam
By 2025-26	10	20%	18	10 MW APDCL solar project under CPSU Ph-II scheme	Solar	APDCL	https://www.mercomindia.com/assam-discom-for-10-mw-solar-project
By 2025-26	50	20%	88	50 MW APDCL solar project on PPP basis	Solar	APDCL	https://www.mercomindia.com/assam-discom-issues-rfs-to-procure-50-mw-solar-power
By 2025-26	70	20%	123	70 MW APDCL solar project on PPP basis	Solar	APDCL	https://www.mercomindia.com/assam-discom-bids-70-mw-solar
By 2025-26	200	20%	350	200 MW APDCL solar project on PPP basis	Solar	APDCL	https://www.mercomindia.com/assam-discom-procure-200-mw-solar
By 2025-26	620	20%	1086	620 MW cumulative solar projects by JV bw APGCL and Oil India	Solar	APDCL	https://www.mercomindia.com/assam-approves-620-mw-renewable-energy-projects
2026-27	22.5	45%	89	Borpani Middle I SHEP	Hydro	APGCL	Exp commissioning after 2025. Estimated project cost of Rs 261.51 Cr
2026-27	60	45%	237	Borpani Upper Stage SHEP	Hydro	APGCL	Exp commissioning after 2025.
Beyond FY 32	660	65%	3570	Margherita Thermal Power Project	Thermal	APGCL	Currently facing environmental issues; Expected to be commissioned beyond FY 32

Details of RE plants in Assam

#	Plant Name	Capacity breakup	Total Capacity (MW)	Type of resource	Type of ownership
1	Champavati / HHPCPL	2*2.025	4.05	Small Hydro	Pvt.
2	Myntreng	3*3+3*1.5	13.5	Small Hydro	State
3	Azure Solar_Boko	1*25	25	Solar	Pvt.
4	Azure Solar_Nagaon	1*15	15	Solar	Pvt.
5	Azure Solar_Pailapool	1*25	25	Solar	Pvt.
6	Azure Solar_Rowta	1*25	25	Solar	Pvt.
7	CALCOM Solar	1*20	20	Solar	Pvt.
8	Maheshwari Mining Solar	1*10	10	Solar	Pvt.
9	Patanjali Ayurved Solar	1*4	4	Solar	Pvt.
10	Suryatap	1*5	5	Solar	Pvt.
11	Solar park Amguri	1*70	70	Solar	Pvt.
12	Kopili II		25	Hydro	Central
TOTAL			241.55		



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